AIDS Research and Therapy https://doi.org/10.1186/s12981-026-00897-y

Article in Press

# From vertical silos to integrated care: administrative agility and normalization of HIV services in Tamil Nadu, India

Received: 27 February 2026

Accepted: 19 May 2026

Published online: 28 June 2026

**Cite this article as: Rafique E.M., Ruknudin A.M., Bukhari S. *et al.* From vertical silos to integrated care: administrative agility and normalization of HIV services in Tamil Nadu, India. *AIDS Res Ther* (2026). https://doi.org/10.1186/s12981-026-00897-y**

**Essa M. Rafique, Abdul M. Ruknudin, Sadiq Bukhari & Mohammad A. Akbarsha**

# From Vertical Silos to Integrated Care: Administrative Agility and Normalization of HIV Services in Tamil Nadu, India.

**Authors: Essa M Rafique[1], Abdul M Ruknudin[2,3], Sadiq Bukhari[3], Mohammad A Akbarsha[3,4]**

[1]Danalakshmi Srinivasan Medical College, Siruvachur, Perambalur, Tamil Nadu, India

[2]Microbiology and Immunology, School of Medicine, University of Maryland, Baltimore, MD 21201, USA; [Retired]

[3]Jamal Vaccine Research Centre (JVRC), Jamal Mohamed College, Tiruchirappalli, 620020, Tamil Nadu, India

[4]Mahatma Gandhi-Doerenkamp Centre for Alternatives to Animal Experiments, Bharathidasan University, Tiruchirappalli 620024, Tamil Nadu, India

Corresponding Author: Mohammad A. Akbarsha, JVRC, Jamal Mohamed College, Tiruchirappalli, 620020, Tamil Nadu, India. Email: akbar.jvrcjmcdu@gmail.com

.

---

## Abbreviations

**AIDS:** Acquired Immunodeficiency Syndrome
**ANC:** Antenatal Care
**ART:** Antiretroviral Therapy
**ARV:** Antiretroviral (Drug)
**CBO:** Community-Based Organization
**CBT:** Community-Based Testing
**CD4:** Cluster of Differentiation 4 (T-lymphocyte cell count)
**CoE:** Centre of Excellence
**DAART:** Decentralized Antiretroviral Therapy
**EVTH:** Elimination of Vertical Transmission of HIV
**FICTC:** Facility-integrated Counselling and Testing Centre
**Four-S:** A mandatory screening tool assessing four clinical symptoms (cough, fever, weight loss, and night sweats) to facilitate bidirectional TB-HIV referrals
**FSW:** Female Sex Worker
**HRG:** High-Risk Group
**HIV:** Human Immunodeficiency Virus
**HIVST:** HIV Self-Testing

**ICTC:** Integrated Counselling and Testing Centre
**IDU:** Injecting Drug User
**LAC:** Link ART Centre
**LFU:** Loss to Follow-up
**MSM:** Men who have Sex with Men
**NACO:** National AIDS Control Organization
**NACP:** National AIDS Control Programme
**NGO:** Non-Governmental Organization
**Ni-kshay:** A unified digital platform for TB-HIV case notification, nutritional incentive delivery, and integrated treatment follow-up across India's public and private health sectors
**OI:** Opportunistic Infection
**PLHIV:** People Living with HIV
**PPTCT:** Prevention of Parent-to-Child Transmission
**SACS:** State AIDS Control Society
**SRL:** State Reference Laboratory
**TANSACS:** Tamil Nadu State AIDS Control Society
**TG:** Transgender
**TI:** Targeted Intervention
**VCTC:** Voluntary Counseling and Testing Centre (now largely merged into ICTC)
**VL:** Viral Load

---

**Abstract**

**Background:** While India’s HIV epidemic discloses substantial inter-state heterogeneity, Tamil Nadu (TN) achieved one of the country’s earliest and most sustained reversals in transmission. Once a high-burden state, TN successfully decoupled its trajectory from other high-prevalence southern regions. This review examines the 'Tamil Nadu Model,' benchmarking its performance against West Bengal and global high-performers to identify specific drivers of success while comparing it with Karnataka, Maharashtra, and undivided Andhra Pradesh.

**Methods:** We synthesized longitudinal data (2000–2024) from HIV Sentinel Surveillance (HSS), IBBS, NACO Spectrum estimates, and NFHS reports. Using a Strategic Blueprints (SB) framework, we analyzed 13 key epidemiological and health-system indicators, including TB-HIV collaborative care. We compared Tamil Nadu's performance with Karnataka, Maharashtra, undivided Andhra Pradesh, and West Bengal, as well as international benchmarks including Thailand, New South Wales, and KwaZulu-Natal.

**Results:** Tamil Nadu recorded a 94% reduction in annual new infections, with adult prevalence falling from 1.15% (2000) to 0.16% (2024). TN became the first major Indian state to achieve the WHO threshold for eliminating vertical transmission (<5%). Comparative analysis demonstrates that TN's "Administrative Agility" facilitated faster declines than low-intensity states like West Bengal. Furthermore, integrating TB-HIV services via the Ni-kshay platform and "Four-S" screening protocols achieved near-universal status awareness among TB patients, accelerating progress toward UNAIDS 95-95-95 targets.

**Conclusion:** Tamil Nadu's trajectory proves that administrative agility and institutional normalization are more effective than vertically siloed interventions. Its ability to outperform regional peers and global benchmarks underscores the robustness of its Strategic Blueprints. As programs transition toward 2030, integrating HIV care with non-communicable disease and TB management offers a saleable road-map for sustained control. Ultimately, shifting from state-level data to global policy lessons positions the "Tamil Nadu Model" as a transferable framework for ending AIDS in resource-limited settings worldwide.

## 1. Introduction

India's HIV epidemic reveals substantial inter-state heterogeneity, with a small number of states historically accounting for a disproportionate share of the national burden. In the early 2000s, southern and western states such as Tamil Nadu, Maharashtra, Karnataka, and undivided Andhra Pradesh reported higher prevalence rates compared to the national average, reflecting localized transmission dynamics and varying programmatic responses [1,2].



Researchers and policymakers have historically described India's HIV epidemic as a concentrated epidemic, with transmission largely confined to key populations, including female sex workers (FSWs), men who have sex with men (MSM), people who inject drugs (PWID), and their sexual partners. During the early phase of the epidemic in the 1990s and early 2000s, high-prevalence states such as Tamil Nadu, Maharashtra, Karnataka, and undivided Andhra Pradesh accounted for a substantial proportion of the national burden, driven primarily by heterosexual transmission linked to FSW networks and, in certain regions, injection drug use [1,3]. Over time, the scale-up of targeted interventions under the National AIDS Control Programme (NACP), including condom promotion, behavior change communication, harm reduction strategies, and expanded access to HIV testing and antiretroviral therapy (ART), contributed to a significant decline in new infections at the national level [2,4]. Despite this progress, transmission dynamics have remained closely associated with key populations, underscoring the importance of sustained, high-coverage interventions to prevent onward transmission to the general population. This epidemiological context provides the basis for examining how specific states, such as Tamil Nadu, achieved differential epidemic control outcomes.

Globally, the integration of HIV services into broader health systems has emerged as a key strategy for improving coverage, reducing stigma, and sustaining long-term epidemic control. Countries such as Thailand have successfully embedded HIV prevention and treatment within primary healthcare systems, while settings such as New South Wales, Australia, have integrated HIV services with sexual health programs to achieve high testing uptake and rapid treatment initiation [5,6]. In sub-Saharan Africa, integration of HIV services with maternal and child health and tuberculosis programs have expanded access to testing and antiretroviral therapy, particularly in high-burden regions such as KwaZulu-Natal, South Africa [7]. These experiences highlight that transitioning from vertical, stand-alone programs to integrated service delivery models can enhance reach, improve continuity of care, and accelerate population-level impact. Within this global context, Tamil Nadu represents a notable example of how early and deliberate integration can reshape epidemic trajectories.



Tamil Nadu, once classified among the high-prevalence states, has demonstrated one of the most sustained and early declines in HIV prevalence in India. The state’s response evolved from an initial vertical, program-driven approach to a more integrated, systems-oriented model that emphasized administrative flexibility, early decentralization, and normalization of HIV services within the public health system [8]. Over time, this approach enabled the state to achieve high testing coverage, early treatment initiation, and sustained viral suppression among people living with HIV.

This review examines the “Tamil Nadu Model” as a case study in epidemic control, with a focus on understanding how a combination of administrative, programmatic, and epidemiological strategies contributed to decoupling its trajectory from other historically high-burden states. By synthesizing longitudinal data across multiple surveillance platforms and applying a Strategic Blueprints (SB) framework, the study aims to identify key drivers of success and assess their relevance for broader HIV control efforts in India and comparable global settings.

## 2. Methods

We applied a structured secondary data synthesis to multiple datasets to examine HIV trends in Tamil Nadu and its comparator regions. Rather than primary statistical modeling, we conducted longitudinal trend analysis, cross-state comparisons, and indicator-based interpretation. We synthesized data from programmatic reports and epidemiological estimates provided by the National AIDS Control Organization (NACO) and the Tamil Nadu State AIDS Control Society (TANSACS), specifically extracting from annual *Sankalak* reports [9], India HIV Estimates technical reports (2010–2024), and the Strategic Information Management System (SIMS) [10].

To enhance international interpretability, we categorized the extracted variables into two domains: **Epidemic Indicators** and **Service Delivery Indicators**. Epidemic indicators—including adult HIV prevalence, Antenatal Care (ANC) prevalence, key population (FSW, MSM, PWID) prevalence, annual new infections (incidence), and AIDS-related mortality—reflect the trajectory and intensity of the epidemic [13,14]. Service delivery indicators capture the scale and effectiveness of health systems, encompassing HIV testing volumes, ART coverage, viral suppression rates, Prevention of Parent-to-Child Transmission (PPTCT) coverage, ART Center density, and harm reduction service uptake [13,14].

We triangulated these datasets to ensure consistency across surveillance, programmatic, and modeled estimates. HIV Sentinel Surveillance (HSS) provided site-specific prevalence trends, while Integrated Biological and Behavioral Surveillance (IBBS) offered detailed behavioral metrics and program exposure data [10,11]. TANSACS and SIMS data supported the longitudinal tracking of granular operational outcomes, such as mother–baby pair follow-up and linkage-to-care efficiency [12,13,14].

We conducted comparative analyses primarily against the high-burden southern and western states of Karnataka, Andhra Pradesh (undivided), and Maharashtra. We included West Bengal selectively as a low-intensity epidemic comparator to contrast reversal trends in high-burden versus lower-burden contexts [8,15]. We excluded Kerala because, despite its proximity, its epidemic is driven by distinct, migration-linked heterogeneous patterns that lack a matching early prevalence peak, rendering trend-based comparisons structurally inconsistent [16].

We further bench-marked these state-level indicators against national averages and evaluated performance relative to the UNAIDS 95-95-95 targets (95% diagnosed, 95% on treatment, 95% virally suppressed) [17]. Finally, we analyzed the findings through the Strategic Blueprints (SB) framework, a systems-level conceptual model comprising 13 interrelated domains—including administrative autonomy, institutional normalization, and TB-HIV collaborative care. This framework maps observed epidemiological changes to specific structural and programmatic interventions, explaining how policy choices translated into sustained impact. Detailed SB components are outlined elsewhere.

The analytical framework concludes with a three-tier comparative synthesis: transitioning from intra-state longitudinal trends to national bench-marking, and finally distilling global policy lessons by triangulating Tamil Nadu's performance against diverse international epidemic control models.

---

## 3. Results and Observations

### 3.1. Epidemiological Trends in Tamil Nadu

#### 3.1.1. HIV prevalence

HIV prevalence in Tamil Nadu peaked in the early 2000s and then declined steadily (Figure 1).

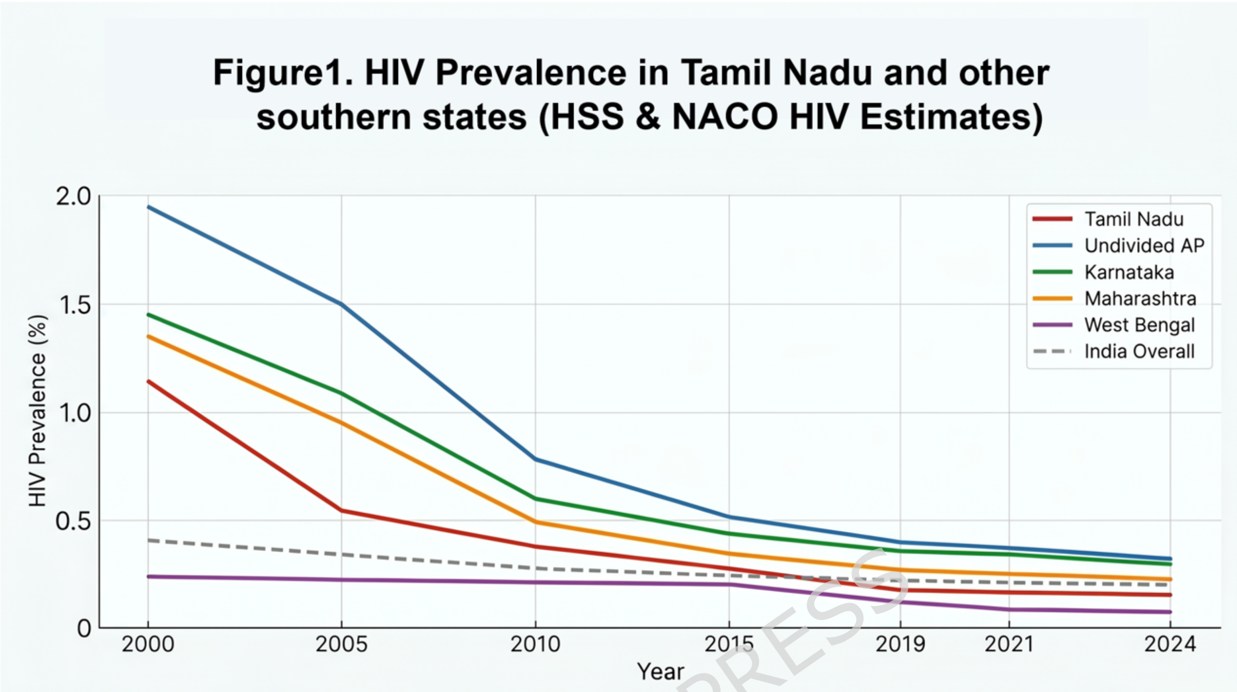


**Figure 1: Trend of HIV Prevalence showing steepest decline in TN when compared to other Southern States 1999-2024 (HSS and NACO HIV Estimates)**

*Note: Data prior to 1999 are excluded to maintain consistency with the standardized National Sentinel Surveillance methodology established under NACP-II. Tamil Nadu's lower baseline prevalence at the start of this period is attributed to the 'Early Mover Advantage'—specifically the 1994 transition to an autonomous state AIDS society (TANSACS), which enabled the state to stabilize transmission in key populations and normalize ANC screening before the national epidemic reached its 2000-2002 peak.*



By 2024 TN recorded one of the steepest and most sustained declines for large Indian states, as adult HIV prevalence fell to approximately 0.16 per cent [18]. This trajectory reflects early strategic decisions that allowed the state to move ahead of the national programmatic timeline.

Although clinicians detected the first clinical case of HIV in India in Tamil Nadu in 1986, standardized state-wide epidemiological tracking only commenced following the launch of the National AIDS Control Programme Phase I (NACP-I) in 1992. Figure 1 initiates its analysis from 1999/2000 to ensure data consistency, as this period marks the transition to the National Sentinel Surveillance system, which provided the first robust, comparable longitudinal datasets across Indian states.

Tamil Nadu's lower prevalence relative to other southern states from 1999 onwards is attributed to its proactive rather than reactive governance. While other regions were still scaling up vertical programs, Tamil Nadu had already established TANSACS in 1994 [19]. This early administrative autonomy allowed the state to saturate high-risk groups with targeted interventions (TIs) and integrate HIV screening into the Primary Health Center (PHC) network nearly five years ahead of the national curve, effectively 'blunting' the epidemic peak before it could reach the levels seen in neighboring states.

The state also demonstrated exceptional outbreak responsiveness. The rapid containment of a severe injecting drug user (IDU) epidemic in Chennai, where prevalence had approached 30 per cent, exemplifies how effectively early harm-reduction strategies work [20]. Collectively, these interventions positioned Tamil Nadu as a national leader in reducing mortality and preventing vertical transmission.

### 3.2. Policy and Programmatic Interventions

#### 3.2.1. Targeted Interventions among Female Sex Workers (FSW)

Tamil Nadu ranked among the earliest states to scale up targeted interventions for female sex workers (FSW), men who have sex with men (MSM), transgender persons, injecting drug users (IDU), migrants, and truckers. FSW HIV prevalence declined sharply from 8.8 per cent in 2003 to below 3 per cent by 2011.

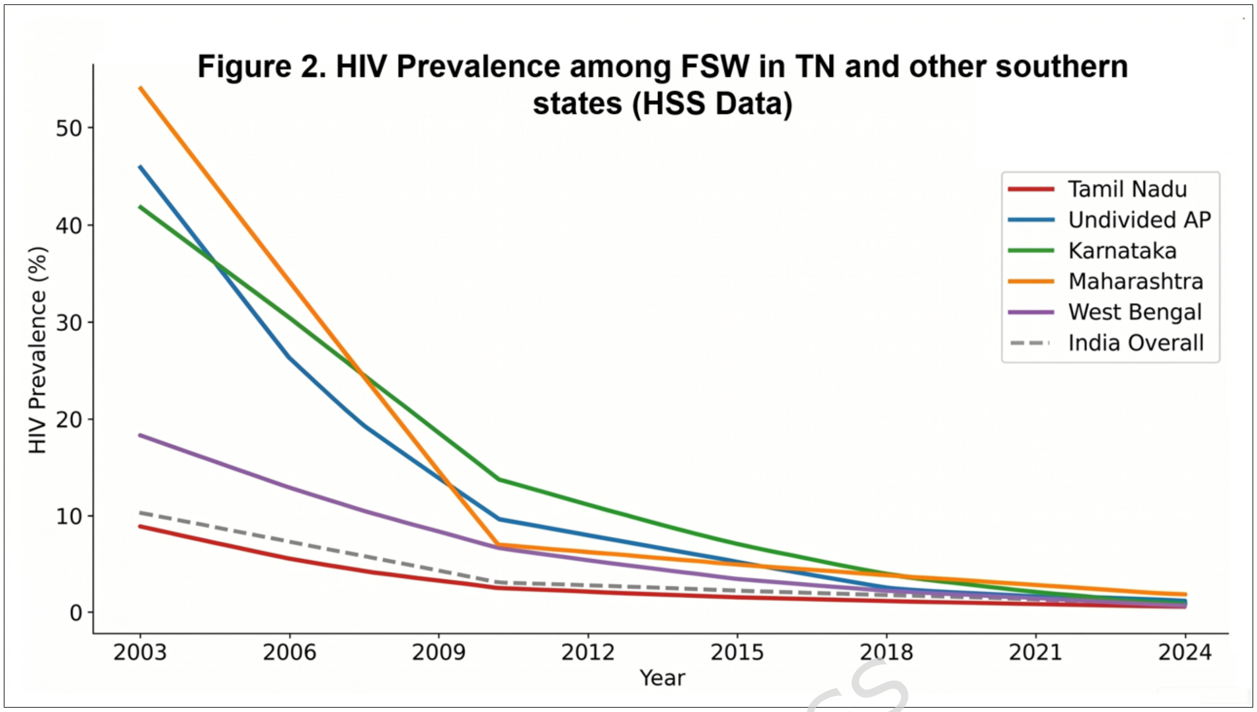


**Figure 2: HIV prevalence among FSW in TN significantly lower than other southern states (HSS data & NACO HIV Estimates)**

Figure 2 show that Tamil Nadu achieved this decline well ahead of Maharashtra and Karnataka [21]. Comparatively, in the early 2000s, Maharashtra was at 54.29% and Andhra Pradesh at 20%, while Tamil Nadu had already stabilized at 8.8%. This highlights that FSW prevalence in Tamil Nadu has continued to decline, remaining consistently below the national average and significantly lower than neighboring high-burden states. Rapid saturation of interventions, decentralized monitoring, and a deliberate shift toward peer-led outreach models drove this early reversal [22].

From 2011 to 2024, HIV prevalence among Female Sex Workers (FSW) entered a 'Maintenance and Surveillance' phase, stabilizing below 2%. This sustained low prevalence in high-risk networks effectively acted as a biomedical firebreak, preventing onward transmission to the general population and directly correlating with the continued decline in Antenatal Clinic (ANC) prevalence rates observed during this period [21, 22, 23].

### 3.2.2. Antenatal Trends

Trends among antenatal clinic attendees (ANC), which serve as a proxy for general population (GP) transmission, underscore the impact of TN’s normalization strategy (Table 1). West Bengal serves as a low-prevalence comparator to contextualize Tamil Nadu's decline against a structurally distinct epidemic baseline.

**Table 1: HIV Prevalence among ANC in TN and other southern states (HSS data & NACO HIV Estimates)**

| ANC Prevalence | | | | | | |
|---|---|---|---|---|---|---|
| **Year** | **Tamil Nadu** | **Karnataka** | **Undivided AP** | **Maharashtra** | **West Bengal** | **India Overall** |
| **2002** | 1.13 | 1.25 | 2.00 | 1.25 | 0.50 | 0.44 |
| **2003** | 0.75 | 1.25 | 1.25 | 0.88 | 0.40 | 0.44 |
| **2004** | 0.63 | 1.18 | 1.15 | 0.75 | 0.45 | 0.39 |
| **2005** | 0.50 | 1.25 | 1.35 | 0.88 | 0.44 | 0.41 |
| **2006** | 0.38 | 0.82 | 1.00 | 0.63 | 0.36 | 0.41 |
| **2007** | 0.35 | 0.75 | 1.00 | 0.50 | 0.38 | 0.48 |
| **2009** | 0.33 | 0.63 | 0.90 | 0.48 | 0.30 | 0.48 |
| **2011** | 0.30 | 0.53 | 0.75 | 0.40 | 0.28 | 0.4 |
| **2013** | 0.28 | 0.50 | 0.59 | 0.38 | 0.25 | 0.35 |
| **2015** | 0.25 | 0.45 | 0.49 | 0.35 | 0.24 | 0.29 |
| **2017** | 0.27 | 0.41 | 0.44 | 0.31 | 0.22 | 0.28 |
| **2019** | 0.18 | 0.38 | 0.40 | 0.28 | 0.21 | 0.24 |
| **2021** | 0.16 | 0.35 | 0.38 | 0.26 | 0.20 | 0.22 |
| **2024** | 0.14 | 0.32 | 0.33 | 0.22 | 0.17 | 0.20 |

The ANC prevalence in the state decreased to below 0.5 per cent by 2005 [23], outperforming other high-burden states by several years. Tamil Nadu achieved this success by embedding HIV testing into routine maternal care by 2004, rather than delivering it as a specialized or stigmatized service [24]. The uniform statewide roll-out of PPTCT services ensured geographic equity and minimized missed opportunities for diagnosis [25].

The early decline in ANC prevalence closely mirrors the decrease that researchers observed among FSWs, supporting the role of targeted interventions in interrupting transmission from core groups to the general population.

### 3.2.3. Concentrated Epidemics Among MSM and IDU

Tamil Nadu circumvented the explosive growth of HIV among MSM that several neighboring states experienced (Figure 3).

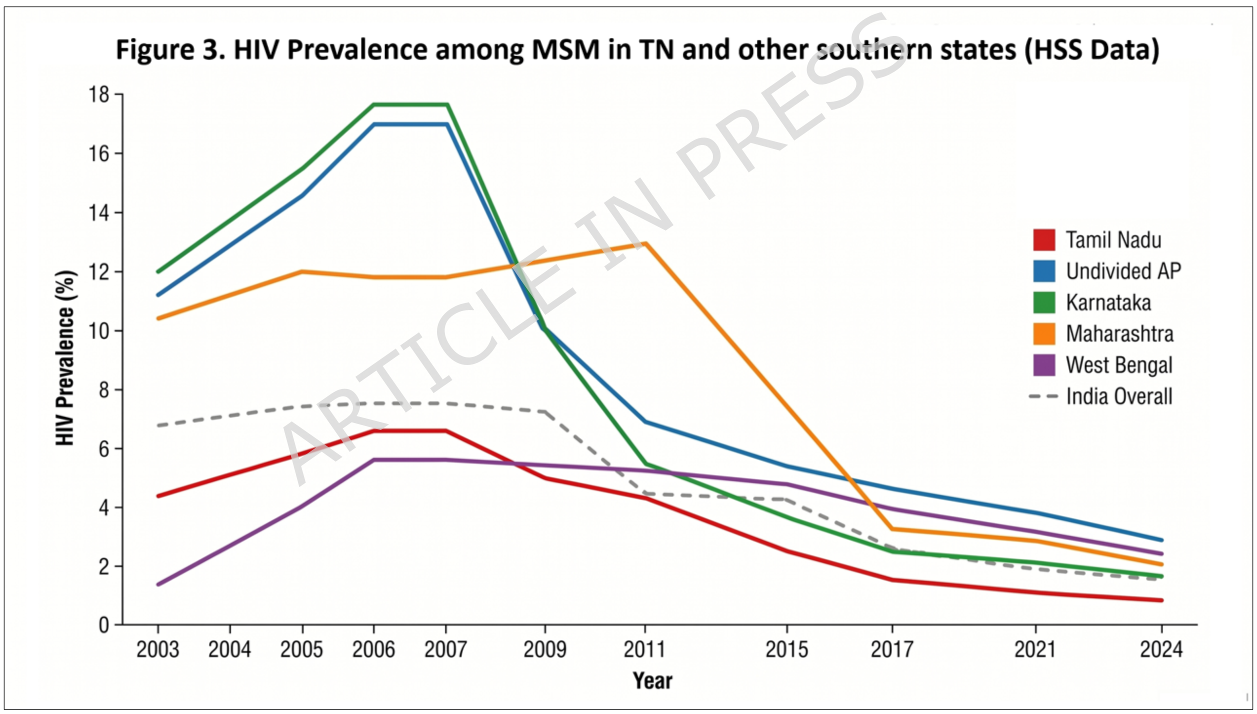


**Figure 3: HIV prevalence among MSM in TN and other southern states as compared with West Bengal (HSS data)**

Early partnerships with community-based organizations, including MSM-led networks, fostered trust, reduced stigma, and facilitated sustained behavioral change. In contrast, states that delayed the expansion of targeted interventions until after prevalence peaked did experience higher cumulative disease burdens [26].

NACO introduced through the National AIDS Control Programme during NACP-III (2007–2012), Opioid substitution therapy (OST) for people who inject drugs (IDU) in India with pilot implementation beginning around 2005–06. However, NACO initiated harm-reduction approaches including needle–syringe exchange programs (NSEP) earlier in the late 1990s under NACP-II (1999–2006), contributing to subsequent declines in HIV prevalence among IDU populations [27].

The state's response to the IDU epidemic represents a global benchmark. By prioritizing needle–syringe exchange programs and Opioid substitution therapy over punitive approaches, Tamil Nadu rapidly reversed one of the highest recorded IDU prevalence rates (Figure 4) [28].

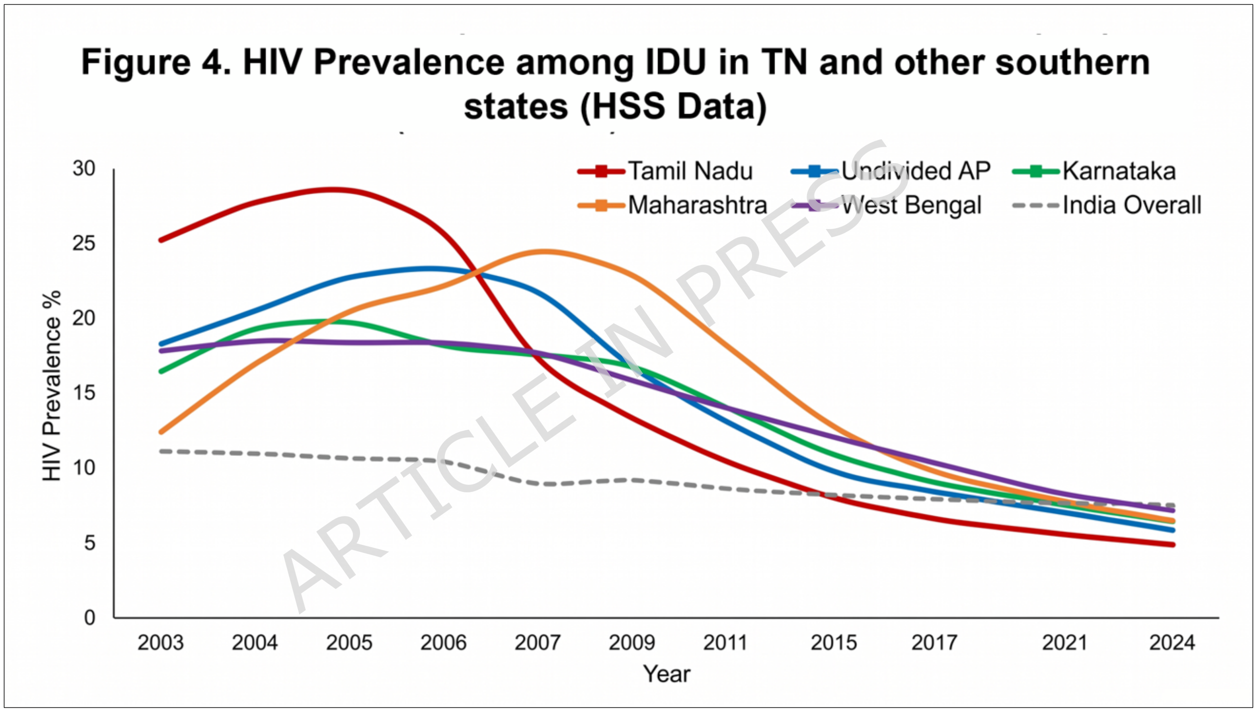


**Figure 4: HIV prevalence among IDU was reduced by over 75% in TN and Maharashtra (HSS data)**

NGO-led community surveillance, improved data quality, and real-time programmatic adjustments reduced prevalence by over 75% from the respective peaks in Tamil Nadu and Maharashtra [29].

Unlike Tamil Nadu, which peaked in 2005 or Maharashtra, which peaked in 2007, West Bengal's IDU prevalence rose significantly later, peaking around 2007–2009.

### 3.3. Mortality Transition and Treatment Outcomes

Annual AIDS-related deaths declined more rapidly in Tamil Nadu than in other high-burden states (Table 2) [30].

**Table 2: AIDS-related deaths in TN and other southern states (HSS data)**

| AIDS-related deaths (Deaths per 100,000 Population) | | | | | | |
|---|---|---|---|---|---|---|
| Year | Tamil Nadu | Karnataka | Undivided AP | Maharashtra | West Bengal | India Overall |
| **2005** | 33.95 | 50.61 | 56.81 | 37.44 | 8.03 | 13.62 |
| **2006** | 30.57 | 46.39 | 52.96 | 34.36 | 7.56 | 12.77 |
| **2007** | 27.29 | 42.30 | 49.19 | 31.37 | 7.09 | 11.95 |
| **2008** | 24.11 | 38.32 | 45.49 | 28.47 | 6.64 | 11.16 |
| **2009** | 21.01 | 34.45 | 41.87 | 25.65 | 6.20 | 10.39 |
| **2010** | 18.00 | 30.70 | 38.32 | 22.91 | 5.77 | 9.64 |
| **2011** | 15.89 | 27.23 | 34.73 | 20.41 | 5.45 | 8.62 |
| **2012** | 13.96 | 23.98 | 31.31 | 18.04 | 5.18 | 7.66 |
| **2013** | 12.05 | 20.78 | 27.94 | 15.71 | 4.90 | 6.71 |
| **2014** | 10.16 | 17.64 | 24.61 | 13.43 | 4.63 | 5.79 |
| **2015** | 8.28 | 14.54 | 21.32 | 11.20 | 4.37 | 4.89 |
| **2016** | 7.36 | 13.09 | 18.73 | 9.86 | 4.02 | 4.44 |
| **2017** | 6.45 | 11.66 | 16.17 | 8.55 | 3.68 | 4.01 |
| **2018** | 5.65 | 10.25 | 14.51 | 7.59 | 3.45 | 3.68 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **2019** | 4.86 | 8.87 | 12.88 | 6.66 | 3.22 | 3.35 |
| **2020** | 4.27 | 7.89 | 11.88 | 6.11 | 2.94 | 3.22 |
| **2021** | 3.69 | 6.92 | 10.89 | 5.57 | 2.66 | 3.09 |
| **2022** | 3.47 | 6.56 | 10.16 | 5.36 | 2.49 | 3.01 |
| **2023** | 3.26 | 6.22 | 9.44 | 5.15 | 2.32 | 2.92 |

In 2005, India reached the peak of pre-ART mortality, but Tamil Nadu leveraged its existing hospital infrastructure to integrate ART services with exceptional speed. By achieving early linkage to care and decentralizing service delivery, the state effectively minimized delays between HIV diagnosis and treatment initiation.

Between 2005 and 2023, Tamil Nadu achieved a profound reduction in AIDS-related mortality. After adjusting for population, the mortality rate plummeted from a peak of 33.95 per 100,000 population in 2005 to 3.26 per 100,000 by 2023 (Table 2). This represents a nearly 90% decline in the mortality rate, significantly outpacing the national reduction trajectory.

By 2023, Tamil Nadu's mortality rate (3.26 per 100,000) remained substantially lower than neighbouring high-burden states such as undivided Andhra Pradesh (9.44 per 100,000) and Karnataka (6.22 per 100,000). Notably, Tamil Nadu's performance has converged with mortality levels traditionally seen in low-prevalence states like West Bengal (2.32 per 100,000) [31]. This accelerated mortality transition reflects the synergistic effects of early prevalence control, efficient decentralized linkage-to-care systems, and the state's success in maintaining high long-term treatment retention.

### 3.4. Incidence Reduction and Prevention Impact

The decline in HIV incidence represents the ultimate measure of prevention success. Tamil Nadu achieved an estimated 94 percent reduction in annual new infections and effectively decoupled its trajectory from the regional average (Figure 5) [32].

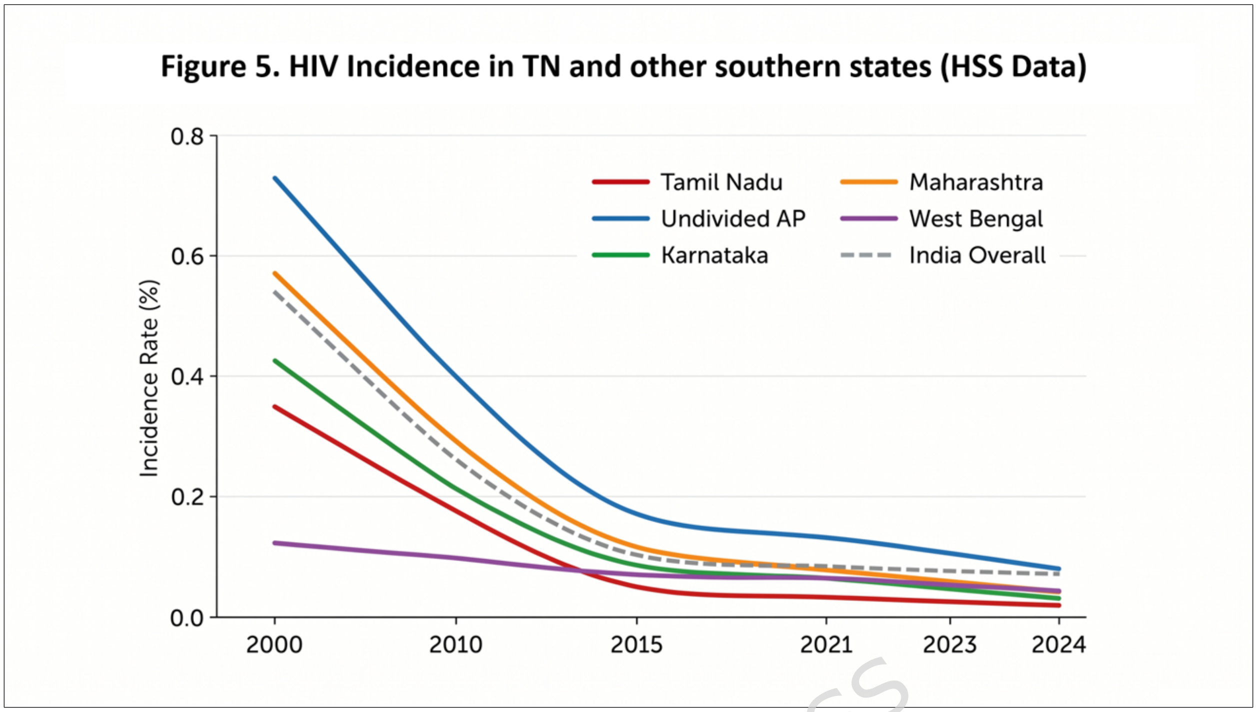


**Figure 5: Data showing that Tamil Nadu remains the "gold standard" for incidence reduction, sitting significantly below the national average (0.02 vs 0.05). (HSS data 1999 -2024)**



Tamil Nadu remains the "gold standard" for incidence reduction, sitting significantly below the national average (0.02 vs 0.05). Tamil Nadu created an early saturation of targeted interventions, which formed a prevention “fire-break,” while administrative autonomy enabled rapid adoption of community-based screening and digital hot-spot mapping [33]. India’s national incidence has stabilized at 0.05 per 1,000, which is a massive achievement from the 2000 peak of 0.54. Thus, India is effectively at 1/10th of the infection velocity compared to prevalence at the turn of the century. Despite massive interventions, the combined Andhra/Telangana region still records an incidence (0.08) higher than the national average, highlighting a need for continued focus on "hot spot" districts there. Maharashtra and West Bengal showed slower declines, which reflect the challenges of urban complexity and low-intensity epidemics, respectively [26]. Tamil Nadu’s continued focus on precision prevention under NACP-V positions it strongly for achieving the 2030 elimination targets [34].

### 3.5. Health System Integration in ART Service Delivery

Tamil Nadu’s Antiviral therapy (ART) infrastructure exemplifies the principle of vertical quality through horizontal integration. Accordingly,

Figure 6 shows the density of ART centers (per 1,000,000 people) for the southern and comparison states from 2004 to 2024 (Figure 6).

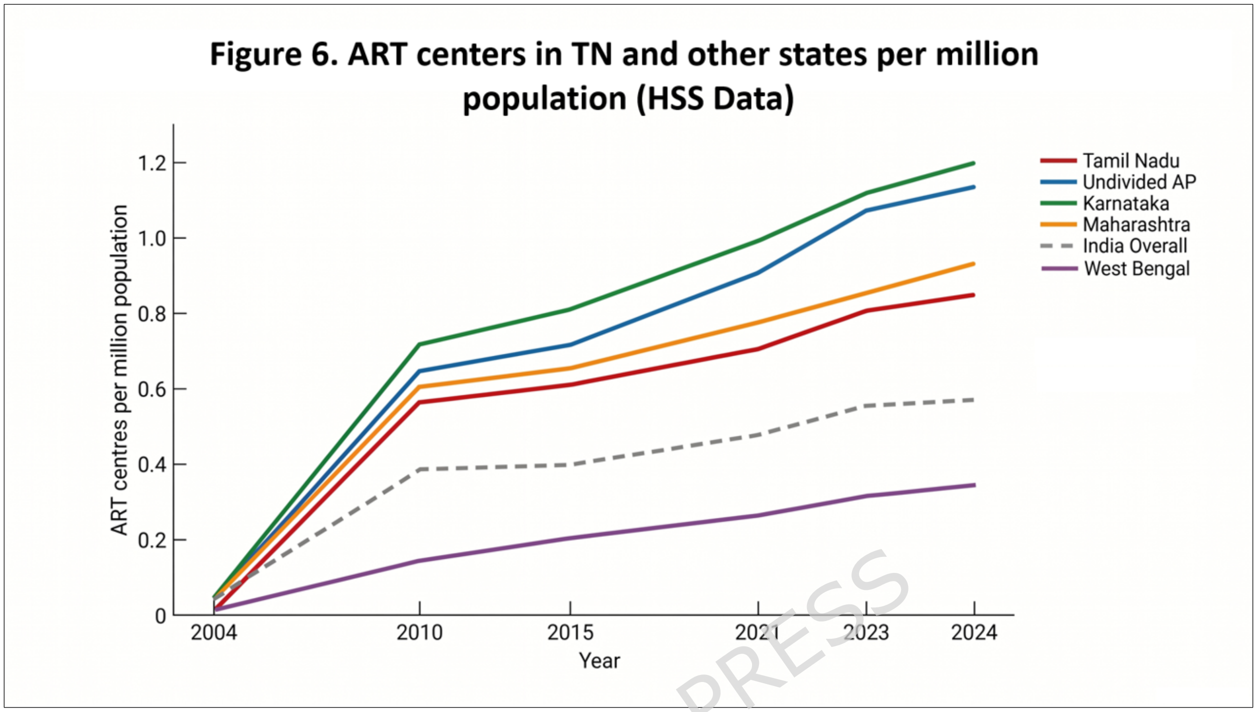


**Figure 6: Data showing that by 2015, TN (0.80) is nearly 50% more "saturated" with ART centers than the national average (HSS and Census data)**



While Maharashtra and Karnataka have higher absolute numbers of ART centers, Karnataka has the highest density per million population (1.079). Tamil Nadu (0.797) shows a highly stable growth in density, indicating that the expansion of the ART network has consistently outpaced its population growth, ensuring sustained coverage. Also, we see that Tamil Nadu, Karnataka, and Undivided AP all crossed the 0.5 centers/million threshold as early as 2010. This explains why the death rates in these states began to plummet during that decade— as the treatment was physically closer to the people

The growth rate in Tamil Nadu slowed down after 2015. This is not a failure, rather it is saturation. For, once we have a center within a reasonable distance of every patient, addition of more would provide diminishing returns. Further, the national average currently remains at ~0.54 centers per million. Tamil Nadu (0.80) is nearly 50% more "saturated" with treatment centers than the national average, which supports the argument about "efficient linkage-to-care."

Thus, by embedding ART centers within government medical colleges and district hospitals, the state normalized HIV care and built patient trust. The subsequent expansion of over 150 Link ART Centers minimized travel-related attrition and improved treatment retention [35]. A tiered system, comprising Centers of Excellence, ART Plus Centers, and Link ART Centers, facilitated differentiated service delivery and used specialized resources efficiently. This proximity-based planning ensured equitable access across rural and urban settings.

### 3.6. Viral Load Monitoring and the Third 95

Tamil Nadu rapidly scaled up viral load testing following the national transition in 2018 (Figure 7).

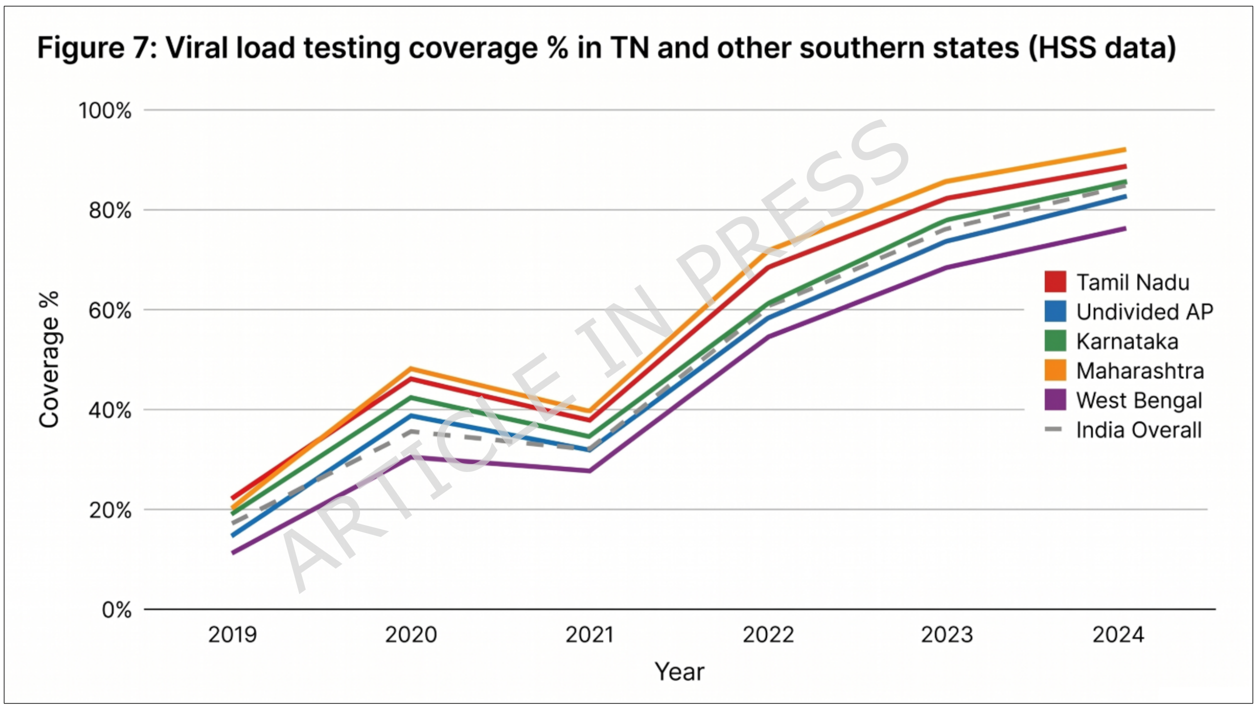


**Figure 7: Data showing that TN achieved the Third 95 goal by ensuring that 95% of all people receiving antiretroviral therapy achieve viral suppression (HSS data)**

A hub-and-spoke laboratory network, anchored by Link ART Centers, provided equitable access to molecular monitoring. However, the year 2021 data clearly illustrates the "COVID-19 Pandemic Pivot” wherein coverage dropped nationally. However, Tamil Nadu's recovery to 82% by 2023 demonstrates a resilient laboratory "Hub and Spoke" model. By 2023–24, viral load coverage reached approximately 88 per cent, while viral suppression rates remained consistently between 92 and 95 per cent [36].

The timely return of results powered rapid clinical decision-making and prevented drug resistance, reinforcing the state's leadership in achieving the Third 95 goal by ensuring that 95% of all people receiving antiretroviral therapy achieve viral suppression [17].

### 3.7. HIV Testing Coverage Outcomes

The routine integration of HIV testing into Primary health care (PHC) services produced near-universal testing coverage among pregnant women [37]. We used HIV testing data from antenatal care (ANC) settings as a proxy to estimate testing coverage in the general population. (Figure 8)

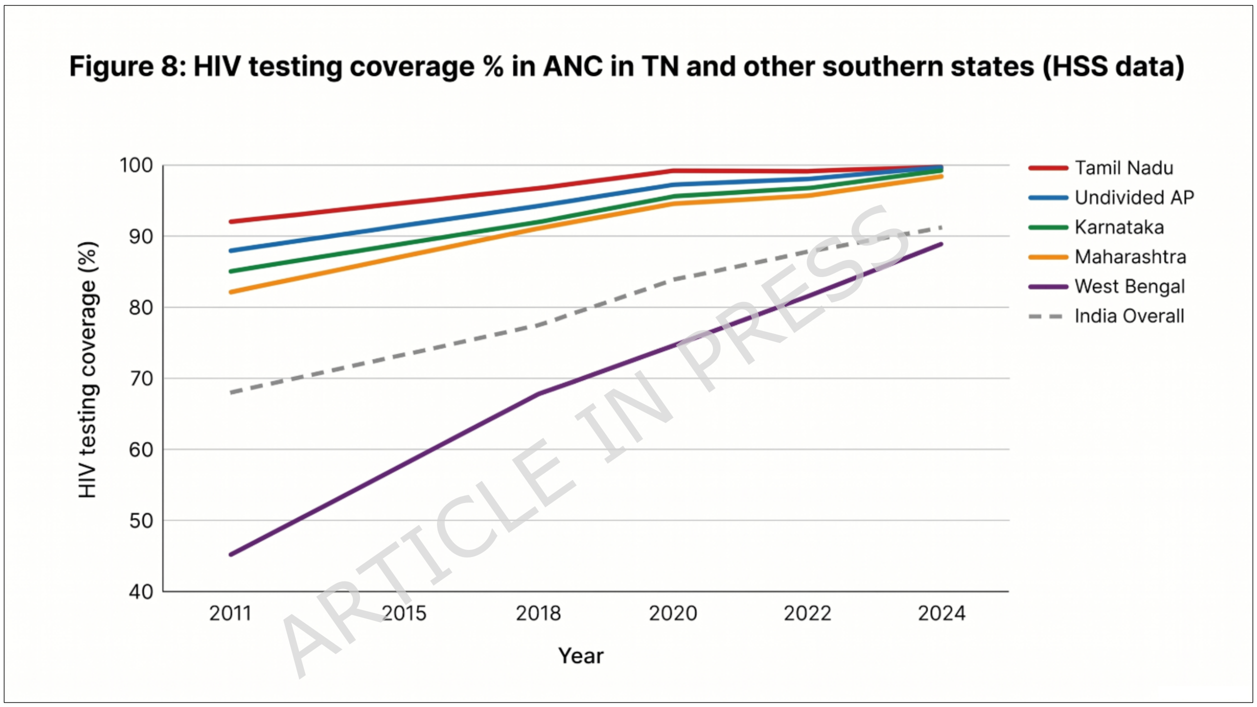


**Figure 8: Data showing that Tamil Nadu reached the 92% coverage mark by 2011 for HIV testing coverage % among ANC (HSS data)**

Tamil Nadu reached 92% coverage as early as 2011, while the national average was still struggling at 68%. By integrating HIV testing into routine Primary Health Centre (PHC) diagnostics (Institutional Normalization), Tamil Nadu achieved near-universal testing coverage much earlier than other states, making this metric a point of pride for the "Tamil Nadu Model". On the other hand, West Bengal, while catching up, traditionally focused more on high-risk groups, leading to lower general population coverage in the early 2010s.Thus, the 24-point lead by TN, underscores the "prevention firebreak" strategy of identifying cases in the general population before they can spread further. Consequently, by 2024, almost all high-prevalence states have converged toward the 98-99% mark. This is a sign of a mature public health system where HIV testing has been fully "normalized" into the standard diagnostic package [37].

### 3.8. PPTCT Coverage Outcomes

While clinical protocols were initiated early, the five-year trajectory (2004–2009) to reduce vertical transmission below the 5% threshold reflected a deliberate transition from vertical, facility-based silos to a horizontal, 'normalized' integration within the state’s Maternal and Child Health (MCH) grid. This period was essential for scaling rural screening and mobilizing the Village Health Nurse (VHN) network to ensure near-universal coverage [37, 38].

#### 3.8.1. PPTCT Coverage%

Tamil Nadu achieved over 90 per cent Prevention of Parent to Child Transmission (PPTCT) coverage by 2009; nearly a decade ahead of most states (Figure 9) [38].

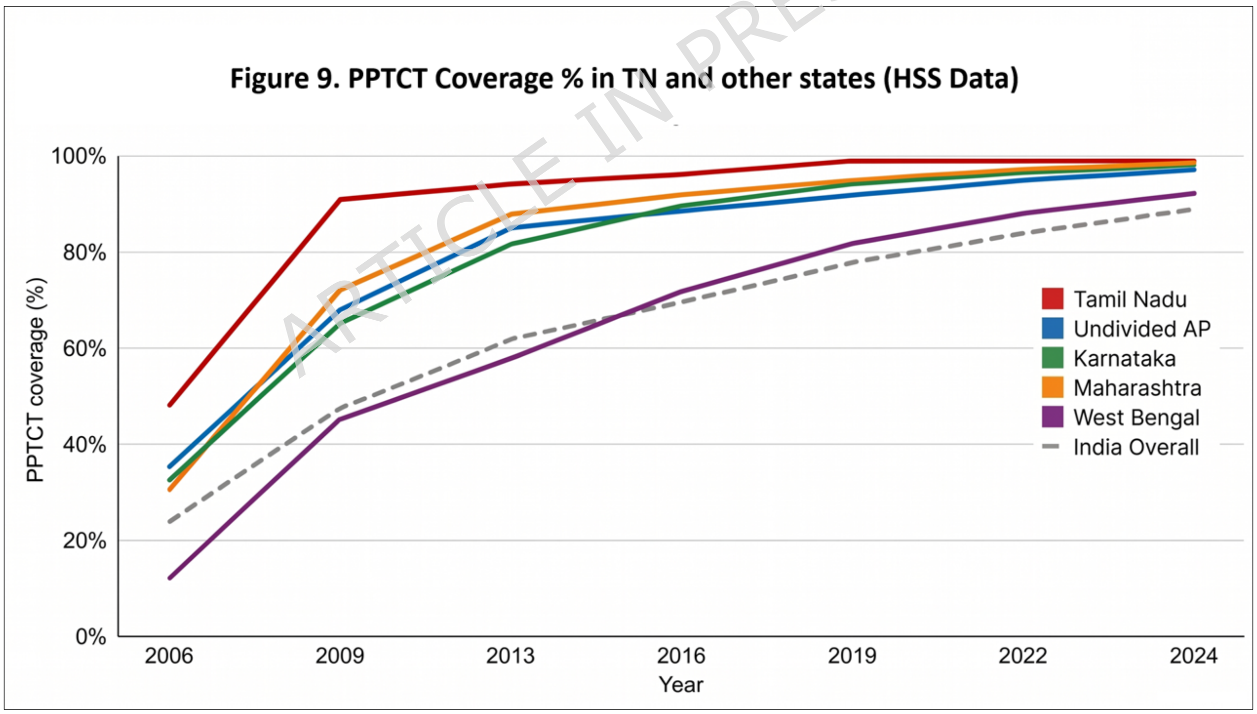


**Figure 9: Data showing Tamil Nadu’s achievement of over 90% Prevention of Parent to Child Transmission (PPTCT) by 2009 (HSS data)**

Tamil Nadu hit 91% coverage, effectively "de-linking" itself from the national trend (which sat at 47%). This was the result of HIV testing becoming an inseparable part of every antenatal checkup. The National Average has climbed steadily but still trails the high-performing southern states by about 10%. This gap represents the "last mile" challenge in states with high home-delivery rates [37]. West Bengal started at a very low baseline (12% in 2006) but has shown consistent, steady growth. WB crossed the 90% threshold only recently, illustrating the time lag in states where MCH-HIV integration was not a first-tier priority in NACP-II [37, 38].

### 3.8.2. **Vertical Transmission Rate**

In TN empowered village health nurses successfully de-stigmatized testing and ensured early antenatal registration. This integrated approach dramatically reduced vertical transmission. Tamil Nadu crossed the WHO elimination threshold of less than 5 per cent vertical transmission around 2018–19 (Figure 10), becoming the first major Indian state to do so [39].

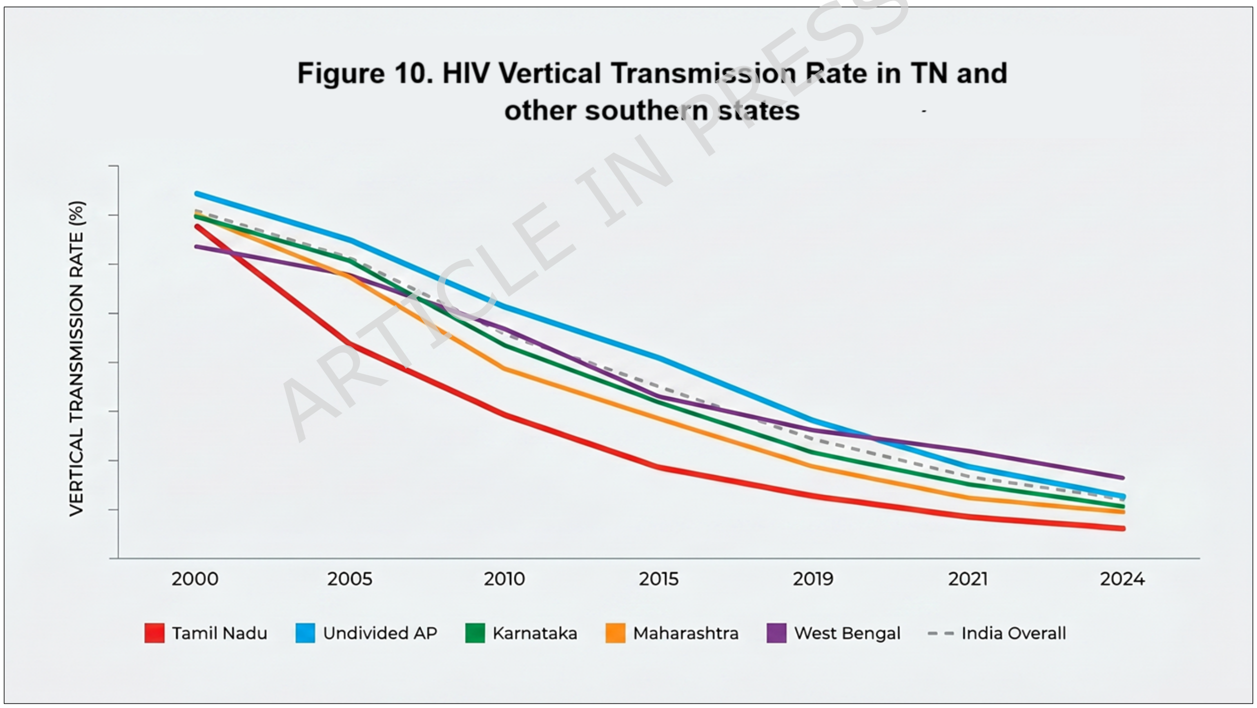


**Figure 10: Tamil Nadu became the first state to cross the WHO threshold of less than 5% vertical transmission rate (HSS data)**

In contrast, we estimated the current national average at 6.1%, meaning the country is still working toward the target that Tamil Nadu achieved half a decade ago. The primary reason for Tamil Nadu’s lead (2.8% vs 6.1% National) is the VHN-led (Village Health Nurse) Maternal Vigilance. By tracking every pregnant woman at the village level, the state ensures that ART initiation happens during the first trimester, which is the single most important factor in reducing transmission to the infant. West Bengal has a higher transmission rate (7.8%) despite having lower general population prevalence. This often indicates a gap in "linkage-to-care"—mothers may be tested, but they are not consistently started on ART early enough in the pregnancy compared to the southern states [37, 39].

### 3.9. TB-HIV Integration: The GHTM Tambaram Model

Tamil Nadu achieved early and near-universal TB-HIV integration by leveraging the historical infrastructure of the Government Hospital of Thoracic Medicine (GHTM) in Tambaram, Chennai. Originally a dedicated tuberculosis sanatorium, GHTM Tambaram evolved into a global referral center for HIV care, effectively "normalizing" the HIV-TB co-infection within a pre-existing specialized pulmonary care framework. This strategic co-location minimized the "referral gap" that often plagues vertical programs. By 2010, Tamil Nadu already initiated 58% of TB-HIV co-infected patients on ART, significantly outperforming the national average of 48% and the 38% seen in West Bengal (Figure 11) [40].



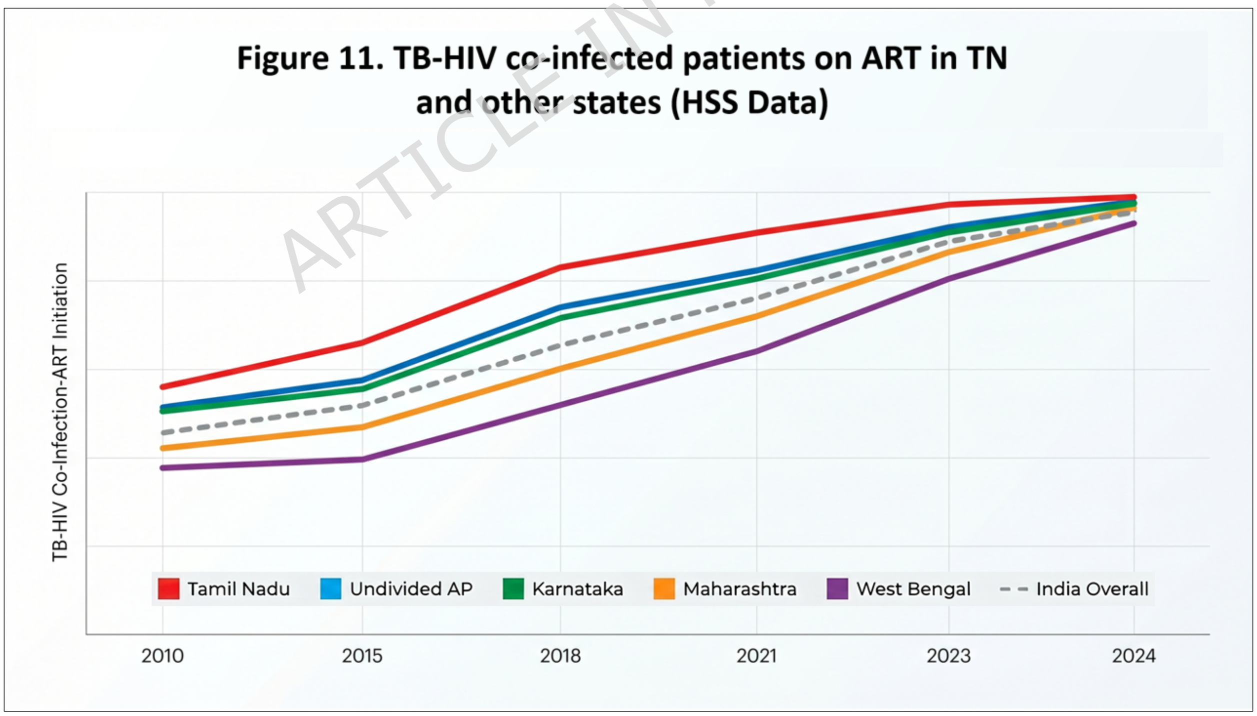


**Figure 11: Tamil Nadu outperforms other states in the TB-HIV integration, which is seen as the "Litmus Test" for health system maturity (HSS data)**

By 2024, Tamil Nadu's integration rate reached 99%, reflecting the successful institutionalization of bidirectional screening—where the state mandates HIV testing for all TB patients and TB screening for all PLHIV [37].

### 3.10. Bench-marking the TN Trajectory Against Global Success Models

To contextualize Tamil Nadu's performance beyond national borders, we bench-marked its epidemiological trajectory against three global high-performing jurisdictions: Thailand (National), New South Wales (Australia), and KwaZulu-Natal (South Africa). These settings represent the three pillars of global HIV response, i) prevention-led, ii) biomedical-led, and iii) treatment-led strategies, respectively.

Rather than a static comparison of prevalence, this exercise evaluates how differing governance structures—ranging from New South Wales' (NSW) clinical precision to Tamil Nadu's autonomous administrative model—successfully collapsed new infection rates and secured long-term survival. As illustrated in Table 3, this cross-jurisdictional analysis tracks progress from the peak of the epidemic (the 1990s/early 2000s) to current 2024/2026 estimates, enabling an assessment of whether Tamil Nadu's trajectory aligns with established global success patterns.



**Table 3: Benchmarking Tamil Nadu's HIV Prevalence Success Against Global Leaders (1990s–2026)**

| Year | Tamil Nadu (India) | Thailand (National) | New South Wales* (Australia) | KwaZulu-Natal (South Africa) |
|---|---|---|---|---|
| **1995** | **1.10%** (Peak Era) | **2.00%** | **0.25%** | **11.00%** |
| **2005** | **0.55%** (Rapid Fall) | **1.40%** | **0.30%** | **25.00%** |
| **2015** | **0.26%** | **1.10%** | **0.18%** | **27.00%** |
| **2024/26** | **0.18%** | **0.90%** | **<0.05%*** | **18.00%** |
| **Success Type** | **Reversal & Containment** | **Aggressive Prevention** | **Virtual Elimination** | **Treatment-Led Stabilization** |

**NSW has essentially achieved the virtual elimination of HIV transmission among MSM, the first jurisdiction globally to do so.*

### 3.11. Key Differentiators: How TN stands out globally

While the quantitative trends in Table 3 demonstrate success across all four jurisdictions, the "Strategic Blueprints", discussed elsewhere, elucidate the specific mechanisms that allowed Tamil Nadu to thrive in a resource-constrained environment.

- TN vs. Thailand (The 100% Condom Program): Thailand gained global recognition for its "100% Condom Program" [38], yet its success relied heavily on top-down, enforcement-led strategies. In contrast, Tamil Nadu achieved comparable results through a bottom-up, rights-based approach [39] (Blueprint 4: Community Agency in Table 5). By prioritizing community empowerment over heavy-handed policing, TN secured sustainable behavioral change while maintaining the dignity of high-risk populations [24, 40].
- TN vs. KwaZulu-Natal, South Africa (Treatment vs. Prevention): As one of the world's most heavily burdened HIV regions, KwaZulu-Natal (KZN) boasts the world's largest ART program—a milestone in "Treatment Success" [41]. However, KZN has historically struggled to contain new infections [42]. Tamil Nadu represents a more balanced dual-success model, simultaneously collapsing the incidence rate while scaling up universal treatment access [22].
- TN vs. New South Wales (NSW), Australia (Resource Intensity): NSW leads the world in biomedical prevention, specifically through the rapid deployment of Pre-Exposure

Prophylaxis (PrEP), and is the first jurisdiction to virtually eliminate transmission among MSM [43]. However, the NSW model is characterized by exceptionally high per-capita costs [44]. Tamil Nadu's "Low-Cost, High-Impact" blueprint offers a more viable and transferable template for the Global South, proving that epidemic reversal is possible without the prohibitive costs of first-world biomedical interventions [45].

Synthesizing these findings, Table 4 highlights the three 'Global Strategic Blueprints' wherein Tamil Nadu's integrated approach offers distinct advantages and lessons over other internationally recognized success models.

**Table 4: Highlighting the three 'Global Strategic Blueprints' or Global Policy Lessons that offer distinct advantages over internationally recognized success models.**

| Name of (SB) Element | Integration Details | Outcome of Integration | Lesson Learnt |
|---|---|---|---|
| **1. Rights-Based Community Agency (vs. Thailand)** | Shift from Thailand’s "top-down" police-led 100% Condom Program to TN’s peer-led, rights-based outreach (Blueprint 4). | Achievement of comparable epidemic reversal without the social cost of coercion or human rights infringements. | Community Agency is more sustainable than State Enforcement; bottom-up empowerment secures long-term behavioral change in high-risk groups. |
| **2. Dual-Track Prevention & Treatment (vs. South Africa)** | Simultaneous scale-up of aggressive Targeted Interventions (TIs) and universal ART, unlike the treatment-heavy focus in KwaZulu-Natal (KZN). | Collapse of the new infection rate (Incidence) alongside high viral suppression, preventing the "Treatment-only Trap." | Clinical Excellence (Treatment) must be integrated with Structural Prevention to stop the engine of the epidemic; one cannot succeed without the other. |
| **3. Frugal Biomedical Innovation (vs. Australia)** | Adoption of a "Low-Cost, High-Impact" model (Tiered Care/Link ART Centers) as an alternative to New South Wales’ high-cost PrEP and specialized clinic model. | "First-World" reversal trends achieved on a "Developing-World" budget, maximizing the reach of limited public health funds. | Resource Intensity is not a prerequisite for elimination; Administrative Agility and decentralized logistics offer a more transferable model for the Global South. |



Ultimately, the Tamil Nadu experience confirms that achievement of first-world epidemiological outcomes does not require first-world budgets, but rather the strategic transition from vertical, top-down programs to integrated, community-led systems [37]. These three global policy lessons suggest that the 'Tamil Nadu Model' offers a resilient and highly transferable road-map for other resource-limited settings striving for the 2030 goal of ending AIDS as a public health threat.

---

## 4. Discussion

### 4.1. Bench-marking the TN's Trajectory

Tamil Nadu transformed key populations from passive recipients of services into active partners in prevention and, thereby, achieved high condom normalization, frequent testing, and early linkage to care. This approach limited onward transmission and protected the general population, as declining antenatal clinic prevalence rates reflect [22]. While the reduction in HIV prevalence among FSWs from 8.8 per cent in 2003 to below 5 per cent occurred over approximately five years, this trajectory reflects a rapid decline within the context of a mature concentrated epidemic. Importantly, behavioral, and programmatic changes, particularly increased condom use, frequent testing, and early linkage to care, likely reduced HIV incidence well before prevalence declined, given the lagging nature of prevalence as an indicator [46].

Following 2011, the available programmatic and surveillance data indicate that FSW prevalence in Tamil Nadu stabilized at low levels, supported by sustained high coverage of targeted interventions, continued condom normalization, and integration with routine testing and treatment services. Although fewer serial prevalence estimates are available during the later period, the absence of resurgence, alongside continued declines in incidence and mortality, suggests durable epidemic control within this key population [47]. Critically, the temporal alignment between declining FSW prevalence and the early and sustained reduction in antenatal clinic (ANC) prevalence supports a population-level prevention effect. As FSWs represent a core transmission group, early saturation of interventions likely reduced onward transmission to clients and, subsequently, to the general population. Declining ANC prevalence reflects this interruption of transmission chains, observed from the mid-2000s onward, indicating that targeted interventions among FSWs contributed substantially to protecting the general population [48].

The TB-HIV integration is the "Litmus Test" for health system maturity. Herein, the GHTM-Tambaram model demonstrates that co-locating services for common comorbidities reduces patient attrition [40]. Tamil Nadu was the first state to integrate ICMR-NIE predictive software into TB screening. Consequently, the TN integrated HIV - TB model (Figure 12) utilizes the Ni-kshay digital platform for unified TB-HIV notification and nutritional incentive delivery, ensuring seamless follow-up across both programs [49].

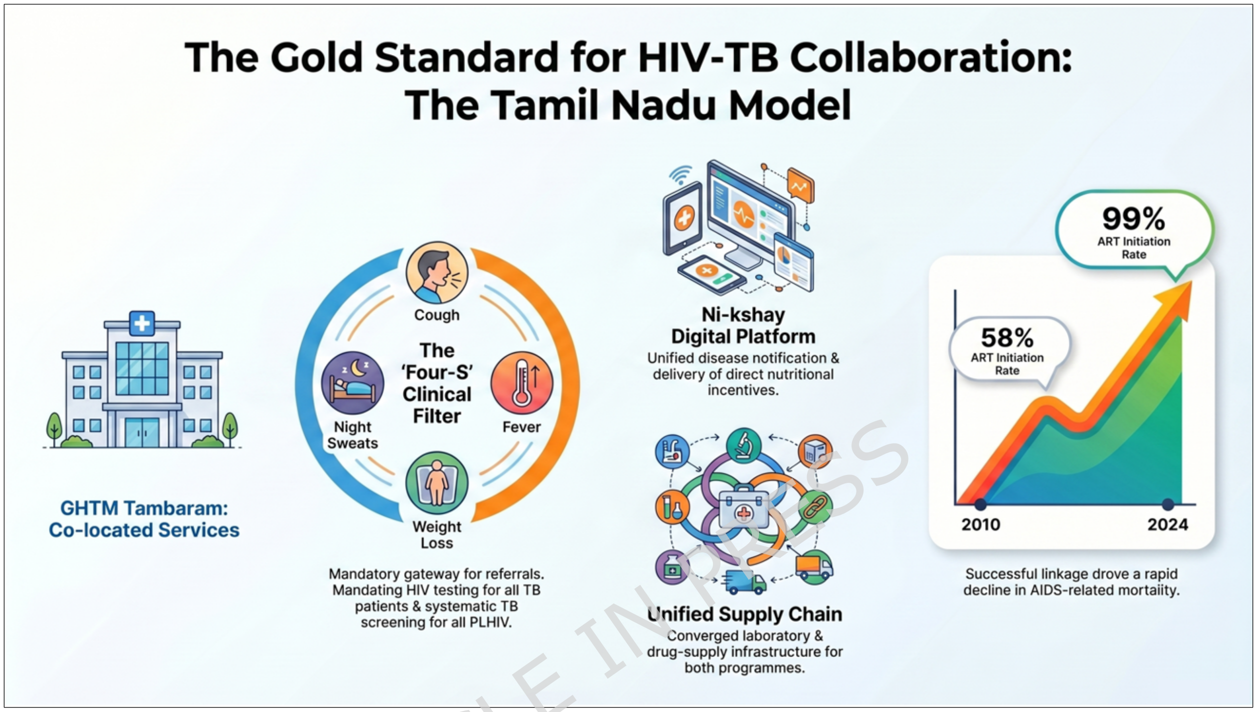


**Figure 12: Tamil Nadu's HIV-TB Model utilizing integrated co-location, and the Ni-kshay digital platform for unified TB-HIV notification and nutritional incentive delivery**

Central to this HIV-TB synergy is the 'Four-S' screening (Cough, Fever, Weight Loss, Night Sweats), which acts as the mandatory clinical gateway for bidirectional referrals between ICTCs and TB clinics [37, 49]. We infer that Tamil Nadu's rapid decline in AIDS-related mortality was partially driven by this high linkage rate, as TB remains the leading opportunistic infection (OI) and cause of death among PLHIV [37, 45]. While West Bengal and other states initially maintained "siloed" (vertical) programs for TB and HIV, Tamil Nadu's early administrative convergence allowed for a unified laboratory and drug-supply chain for both diseases [50]. The transition in Tamil Nadu's TB-HIV Co-infection, with ART Initiation Rate (%) from 58% to 99%, proves that institutionalizing "bidirectional screening" at the Primary Health Centre (PHC) level is more effective than relying on centralized referral hospitals alone [44]. The National Average has caught up significantly since 2018 due to the "National Strategic Plan for TB Elimination," but Tamil Nadu remains the benchmark for nearly perfect linkage (99%) [50].

#### 4.1.1. The Tamil Nadu Model: Strategic Blueprints (SB) for Success

Our analysis shows that Tamil Nadu consistently outperforms national benchmarks across core epidemiological indicators. In 2024, the state achieved an adult HIV prevalence of 0.16% [18], which remains lower than the national average of 0.20% [51]. The state also strengthened the Elimination of Vertical Transmission of HIV (EVTH) program by reducing HIV prevalence among pregnant women to 0.16% from 0.38% in 2010 and by maintaining viral suppression rates above 95% among individuals on treatment. These outcomes exceed the national trajectory toward the UNAIDS 95-95-95 targets [17].

The Targeted Intervention (TI) package for Female Sex Workers (FSW) demonstrates how incorporating Strategic Blueprint elements achieved saturation through a four-pronged approach: (1) peer-led behavioral change communication, (2) standardized condom promotion and distribution, (3) semi-annual syphilis screening with linked HIV testing, and (4) robust treatment of symptomatic sexually transmitted infections (STIs) [4, 8]. Tamil Nadu transitioned FSWs from passive recipients to active partners by institutionalizing the fourth element of the Strategic Blueprints (Community Agency & Ownership, in Table 5) via Community-Based Organizations (CBOs). These CBOs were empowered to manage their own drop-in centers and outreach schedules, effectively shifting the governance of the program to the community. This agency was further solidified through 'Social Protection' initiatives, where CBOs assisted members in accessing legal aid and government social security, thereby building the trust necessary for sustained programmatic engagement [37].

A critical, yet often overlooked, ingredient of the Tamil Nadu success was the bidirectional integration of TB and HIV services. By utilizing the Nikshay digital platform and mandatory 'Four-S' symptom screening at all ART centers, the state achieved near-universal HIV status awareness among TB patients, further reducing co-morbidity-linked mortality [50].

Tamil Nadu achieved these outcomes through deliberate Strategic Blueprints (Table 5) that prioritized administrative autonomy and early decentralized governance rather than relying solely on clinical interventions.

### 4.2. Architecture of SB

The Tamil Nadu model integrates early administrative autonomy, institutional normalization of services, community ownership, harm-reduction policies, decentralized logistics, precision surveillance, and differentiated service delivery [39] (Table 5).

**Table 5: Strategic Blueprints (SB) of the Tamil Nadu HIV Model**

| No. | Name of (SB) Element | Integration Details | Outcome of Integration | Lesson Learnt |
|---|---|---|---|---|
| **1** | Administrative Autonomy (TANSACS) | Tamil Nadu registered TANSACS as an autonomous society (1994) to bypass bureaucratic red tape and fiscal delays. | TANSACS ensured leadership continuity, rapid procurement of life-saving drugs, and faster policy implementation than neighbors. | Structural independence serves as the primary prerequisite for an agile and resilient public health response. |
| **2** | Institutional Normalization | Tamil Nadu integrated HIV testing into the routine diagnostic package of all Primary Health Centers (PHCs). | This integration de-stigmatized the test; general population testing coverage reached >99% by 2020. | Universalizing a service provides the most effective antidote to social stigma and clinical exceptionalities. |
| **3** | Horizontal Infrastructure Hijack | Tamil Nadu leveraged the historically strong Maternal and Child | Tamil Nadu achieved 91% PPTCT coverage by 2009; a decade | Strong pre-existing health systems provide a "ready-made" vehicle for rapid |

| | | | | |
|---|---|---|---|---|
| | | Health (MCH) grid for HIV-specific interventions. | ahead of the national NACP timeline. | infectious disease control. |
| **4** | Community Agency & Ownership: CBO | Tamil Nadu shifted from "targeting" HRGs to "partnering" with community networks (e.g., Sahodaran for MSM, Positive Networks). | This shift truncated prevalence peaks among high-risk groups and secured high long-term treatment retention. | Peer-led models transform patients from passive subjects into active "drivers" of programmatic success. |
| **5** | Harm Reduction as Policy | Tamil Nadu replaced punitive drug measures with intensive Needle-Syringe Exchange (NSEP) and Opioid Substitution Therapy (OST). | Tamil Nadu reversed a world-high 30% IDU prevalence spike in Chennai to stable, manageable levels. | Rights-based, scientific interventions outperform moralistic or punitive approaches in reaching hidden groups. |
| **6** | Decentralized 'Hub & Spoke' Logistics | Tamil Nadu created a dense network of 150+ Link ART Centers (LACs) for medication and decentralized Viral Load (VL) sampling. | Tamil Nadu achieved the fastest "Mortality Transition" in India and high viral suppression rates (92-95%). | Proximity to care determines treatment adherence and lifelong patient survival. |

| | | | | |
|---|---|---|---|---|
| **7** | Precision Hotspot Mapping | Tamil Nadu adopted sentinel surveillance and digital tools (SOCH) early for real-time epidemiological monitoring. | Tamil Nadu identified and contained new clusters within weeks, leading to a 94% collapse in new infections. | Data acts as a "fire-fighting" tool; real-time surveillance must trigger immediate, localized resource shifts. |
| **8** | VHN-Led Maternal Vigilance | Village Health Nurses (VHNs) empowered themselves to facilitate early first-trimester screening and counseling at the household. | VHNs drastically reduced "late presenters" at delivery; Tamil Nadu became the first major state to cross the WHO <5% VTR threshold. | Front-line workers form the most critical link in the chain for the elimination of vertical transmission. |
| **9** | Differentiated Service Delivery | Tamil Nadu implemented a tiered clinical system: Center of Excellence (CoE) for 3rd line ART, ART Plus for 2nd line ART, and Link ART Centers (LACs) for routine refills. | This system optimized utilization of dedicated specialists while reducing the economic burden (travel/lost wages) for stable patients. | A "one-size-fits-all" clinical model proves inefficient; care must stratify based on patient stability. |
| **10** | Closing the Feedback Loop | Tamil Nadu streamlined VL result reporting from central molecular labs back to rural | This streamlining prevented drug-resistant strains and enabled early | Diagnostic technology proves effective only as fast as the subsequent clinical and |

| | | | | |
|---|---|---|---|---|
| | | clinicians for rapid treatment pivots. | identification of treatment failure before clinical symptoms appear. | administrative action. |
| **11** | Early Saturation (Prevention Firebreak) | Tamil Nadu aggressively scaled Targeted Interventions (TIs) for FSWs to near-total coverage as early as 1994. | FSW prevalence fell to 4.0% by 2004 (vs. 41% in Maharashtra), preventing the "bridge" to the general population. | Early saturation of services acts as a "prevention fire-break" that protects the wider public. |
| **12** | Adherence-as-Elimination | Tamil Nadu integrated Viral Load suppression monitoring specifically into the PPTCT cascade to ensure undetectable = un-transmittable ("U=U") before delivery. | Tamil Nadu achieved the <5% Vertical Transmission Milestone (EMTCT) despite the residual risks of breastfeeding. | Eliminating pediatric HIV requires moving beyond "giving a pill" to ensuring molecular-level viral suppression. |
| **13** | TB-HIV Bidirectional Collaborative Care | Co-location of Integrated Counselling and Testing Centres (ICTC) and National TB Elimination Programme | Achievement of near 100% HIV status awareness among TB patients and a significant reduction in | Administrative Agility is maximized when vertical programs share digital surveillance (e.g., Ni-kshay) |

| | | | | |
|---|---|---|---|---|
| | | (NTEP) clinics; implementation of mandatory 'Four-S' symptom screening at all ART centers and GeneXpert diagnostic sharing. | TB-associated mortality among PLHIV through early prophylactic treatment (IPT). | and diagnostic infrastructure, reducing patient "leakage" between specialties. |

Tamil Nadu established the first State AIDS Control Society (TANSACS) in 1994 and created a flexible operational framework that bypassed traditional bureaucratic delays. This framework enabled rapid fund mobilization and "normalization" of HIV services within the existing public health infrastructure [48]. Tamil Nadu transitioned from a vertical, isolated program to an integrated institutional model, which proved essential in de-stigmatizing care. By integrating HIV services, ranging from diagnostics to antiretroviral therapy (ART) with bidirectional HIV-TB collaboration, the state matched the accessibility and routine nature of any other chronic disease management program [19].

These interlocking elements transformed HIV from a specialized crisis into a routine public health service and ensured durable epidemic control. Consequently, Figure 13 delineates the relationship between the three primary Strategic Blueprints and their respective public health achievements.

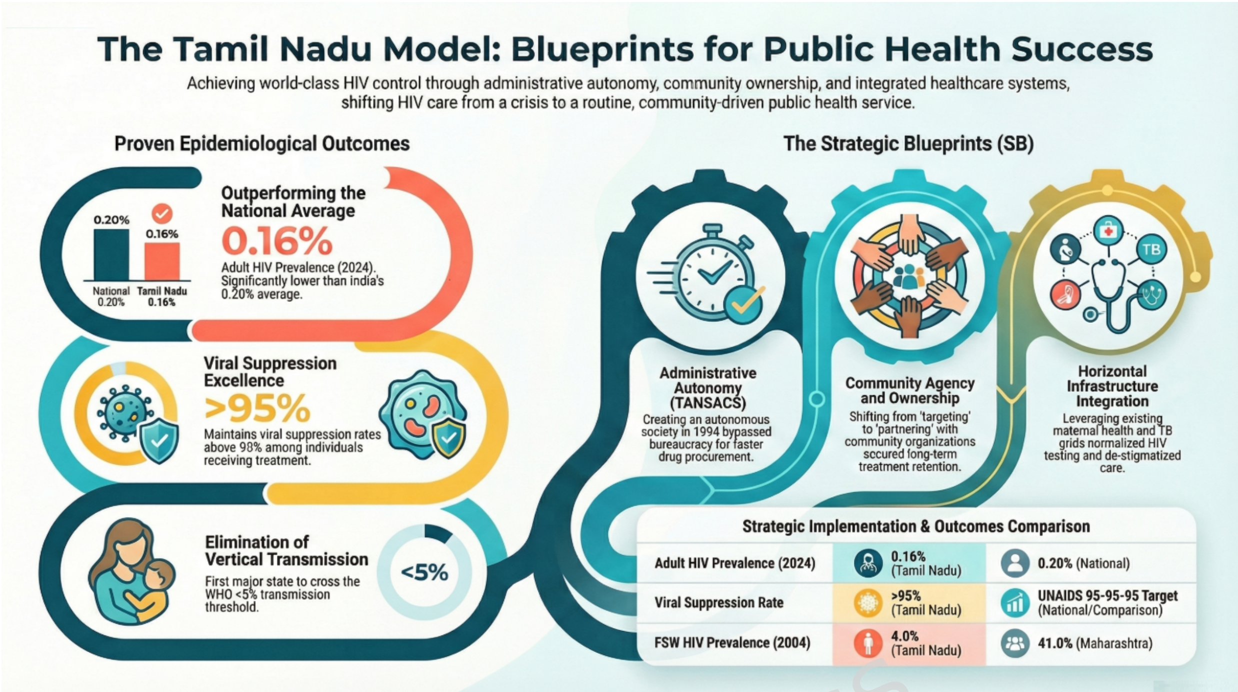


**Figure 13: Showing the three key Strategic Blueprints for Public Health Success and their Proven Epidemiological Outcomes.**



### 4.3. Why TN outperformed globally and locally:

Globally, TN stands out because it achieved "First-World" reversal trends on a "Developing-World" budget. Unlike Thailand, TN avoided coercion; unlike NSW, TN skipped expensive PrEP; and unlike KZN, TN acted before a generalized disaster. Domestically, Tamil Nadu treated national HIV guidelines as a minimum standard, not an upper limit. Early autonomy enabled rapid decision-making, while integration into maternal and primary healthcare normalized services and expanded reach. Peer-led governance reduced stigma [24] and loss to follow-up, and proximity-based logistics maximized adherence [52]. Real-time surveillance ensured rapid containment of emerging clusters. Together, these factors explain the state's outlier performance (Figure 14).

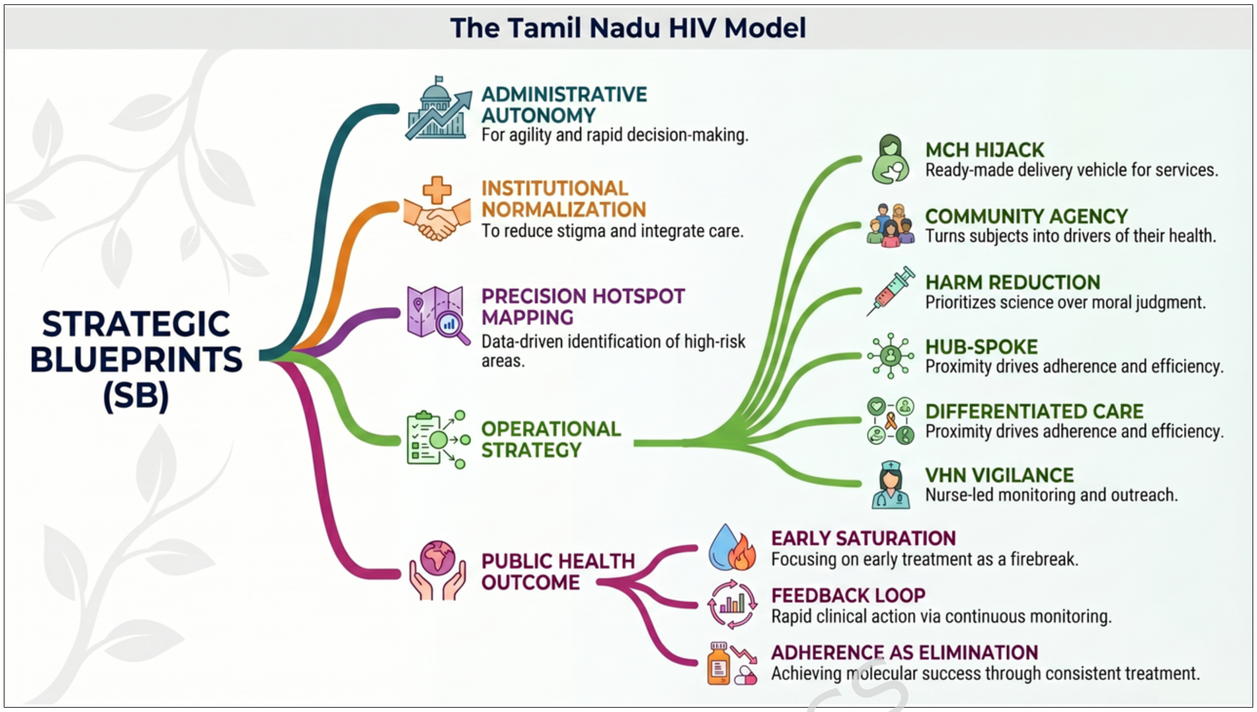


**Figure 14: Strategic Blueprints (SB) that explain the TN HIV model's performance**

### 4.4. Limitations of the Strategic Blueprints Framework:

While the Strategic Blueprints (SB) framework in Table 5 provides a structured approach to linking programmatic interventions with epidemiological outcomes, several limitations must be acknowledged. First, the framework relies on ecological and program-level data, which limits causal inference at the individual level [53, 54]. In addition, unmeasured confounders, like socioeconomic transitions, migration patterns, and broader health system strengthening, may influence the observed associations between interventions and outcomes.

Second, the retrospective synthesis of multiple data sources over a 25-year period introduces potential inconsistencies in measurement methods, surveillance coverage, and reporting accuracy. Although triangulation across HSS, IBBS, NFHS, and NACO estimates improves robustness, we cannot exclude residual bias [11, 55].

Third, the SB framework emphasizes system-level drivers and may under-represent micro-level behavioral heterogeneity within key populations. Aggregated indicators do not fully capture variations in intervention uptake, stigma, and access to care across subgroups [56, 57].

Finally, comparisons with other states and global settings are subject to structural differences in epidemic typology, health system capacity, and data quality. While these comparisons provide useful context, Readers should interpret these comparisons as indicative rather than strictly equivalent [58].

Despite these limitations, the SB framework offers a practical heuristic for understanding how coordinated administrative and programmatic strategies translate into population-level epidemic control.

---

## 5. Future Challenges and Conclusion

As the epidemic matures, Tamil Nadu must care for an aging population of people living with HIV. Tamil Nadu should integrate HIV services with non-communicable disease management, adopt digital targeted interventions for hidden populations [59], and ensure treatment continuity for migrant workers, who often face fragmented care pathways, delayed diagnosis, and barriers to sustained treatment access, as documented in studies from India [60]. Mainstreaming HIV financing into universal health coverage frameworks will sustain gains in a low-incidence era particularly for mobile and underserved populations [61].



Tamil Nadu represents one of India’s most compelling HIV public health success stories. Early action, administrative autonomy, community partnership, and health system integration enabled faster declines in HIV incidence and mortality than in most Indian states [62].

While this review began as an assessment of Tamil Nadu’s success within the Indian context, the evidence suggests that the state’s Strategic Blueprints, Table 5, —centered on administrative agility and institutional normalization—transcend regional boundaries. By successfully decoupling its epidemic trajectory from its high-burden peers on a modest budget, Tamil Nadu provides a 'First-World' result via a 'Global South' methodology. As we look beyond 2026, the transition from vertical silos to integrated primary care stands as the definitive global policy lesson (Table 4) for health systems striving to sustain epidemic control while addressing the emerging burden of non-communicable comorbidities.

**Ethics Approval and Consent to Participate**

This review article did not involve Animal/ Human Experimentation. Therefore, the question of ethics approval does not arise. All Authors and the Institutions concerned have consented to participate in the publication.

**Consent for publication**

All authors consent to this publication.

**Data Availability Statement**

The data that support the findings of this study are available from the corresponding author upon reasonable request.

**Conflict of Interest**

The authors certify that they have no conflict of interest with any financial organization regarding the material discussed in the manuscript.

**Funding**

This research did not receive any external funding.

**Authors' Consent**

All authors read and approved the final version of the manuscript.

**Acknowledgment**

We sincerely thank Dr. H. E. Syed Mohamed, Dr. Mohamed Hussain and Zakeer Hussain, Department of Zoology, Jamal Mohamed College, for their insightful suggestions, and continuous support in this review.



---

## References

[1] National AIDS Control Organization & ICMR-National Institute of Medical Statistics (2019). India HIV Estimates 2019: Fact Sheet. New Delhi: NACO, Ministry of Health and Family Welfare, Government of India. [Online].; 2019 [cited 2025 Dec 31]. Accessed from: https://www.aidsdatahub.org/sites/default/files/resource/india-hiv-estimates-2019.pdf

[2] National AIDS Control Organization & ICMR-National Institute of Medical Statistics (2023). India HIV Estimates 2023: Technical Report. New Delhi: NACO, Ministry of Health and Family Welfare, Government of India. National Institute for Research in Digital Health and Data Science [Online].; 2023 [cited 2025 Dec 31]. Accessed from: https://naco.mohfw.gov.in/sites/default/files/India%20HIV%20Estimates%202023_Technical%20Report_Final_17%20DEC%202024%20(1).pdf#:~:text=HIV%20burden%20estimation%20is%20a%20key%20activity,Dat

a%20Science%20(ICMR%2DNIRDHDS)%20and%20other%20National/Regional%20institutes.

[3] Vickerman P, Foss AM, Pickles M, Deering K, Verma S, Eric Demers, Lowndes CM, Moses S, Alary M, Boily MC. To what extent is the HIV epidemic in southern India driven by commercial sex? A modelling analysis. AIDS. 2010;24[16]:2563-2572. doi: 10.1097/QAD.0b013e32833e8663 [Online].; 2010 [cited 2025 Dec 31]. Accessed from: https://pubmed.ncbi.nlm.nih.gov/20852405/

[4] Dandona, L., Benotsch, E.G. Evaluation of the Avahan HIV prevention initiative in India. *BMC Public Health* 2011;**11**(Suppl 6), 11. https://doi.org/10.1186/1471-2458-11-S6-I1 [Online].; 2011 [cited 2025 Dec 31]. Accessed from: https://link.springer.com/article/10.1186/1471-2458-11-S6-I1

[5] World Health Organization. Consolidated guidelines on HIV prevention, testing, treatment, service delivery, and monitoring: recommendations for a public health approach. Geneva: World Health Organization; 2021. [cited 2025 Dec 31]. Accessed from: https://www.who.int/publications/i/item/9789240031593

[6] Grulich AE, Guy R, Amin J, Jin F, Selvey C, Holden J, et al. Population-level effectiveness of rapid, targeted, high-coverage roll-out of HIV pre-exposure prophylaxis in men who have sex with men: the EPIC-NSW prospective cohort study. *Lancet HIV*. 2018;5(11): e629–e637. [Online].; 2018. [cited 2025 Dec 31]. Accessed from: https://www.sciencedirect.com/science/article/abs/pii/S2352301818302157

[7] Church K, Wringe A, Fakudze P, Kikuvi J, Simelane D, Mayhew SH. Are integrated HIV services less stigmatizing than stand-alone models of care? A comparative case study from Swaziland. BMC Health Serv Res. 2013; 13:403. [Online].; 2018. [cited 2025 Dec 31]. Accessed from: https://pubmed.ncbi.nlm.nih.gov/23336726/

[8] Pickles M, Boily M, Vickerman P et al. Assessment of the population-level effectiveness of the Avahan HIV-prevention programme in South India: a preplanned, causal-pathway-based modelling analysis. The Lancet Global Health, 2013;1, e289-e299. [Online].; 2013 [cited 2025 Dec 31]. Accessed from: https://www.thelancet.com/journals/langlo/article/PIIS2214-109X(13)70083-4/fulltext

[9] Sankalak: Status of National AIDS Response. HIV and AIDS Data Hub for Asia Pacific: Evidence to Action [Online].; 2017 [cited 2025 Dec 31. Accessed from: https://www.aidsdatahub.org/resource/sankalak-status-national-aids-response.

[10] NACO. Strategic Information Management System (SIMS) National Aids Control Organisation. [Online].; 2012 [cited 2025 Dec 31]. Accessed from: https://naco.gov.in/sims.
[11] National AIDS Control Organisation (NACO). A Report on National Data Analysis Plan: A Systematic Approach Towards Analysis of National HIV/AIDS Programme Data. New Delhi: Ministry of Health and Family Welfare, Government of India; 2015. [Online].; 2015 [cited 2025 Dec 31]. Accessed from: https://www.naco.gov.in/sites/default/files/NDAP%20Report%202015.pdf
[12] National AIDS Control Organisation (NACO) Direct Estimation of New HIV Infections Among Key Populations Using Programme Data. Mizoram. Ministry of Health and Family Welfare, Government of India; 2018 [cited 2025 Dec 31]. Accessed from: https://www.naco.gov.in/sites/default/files/Technical%20Brief%20on%20Estimating%20New%20Infections.pdf
[13] National AIDS Control Organization & ICMR-National Institute of Medical Statistics (2022). India HIV Estimates 2021: Fact Sheet. New Delhi: NACO, Ministry of Health and Family Welfare, Government of India. [Online].; 2021 [cited 2025 Dec 31]. Accessed from: https://naco.gov.in/sites/default/files/India%20HIV%20Estimates%202021%20_Fact%20Sheets__Final_Shared_24_08_2022_0.pdf.
[14] National AIDS Control Organisation (NACO) National Data Analysis Plan – Round II, Ministry of Health and Family Welfare, Government of India; 2021 [cited 2025 Dec 31]. Accessed from: https://naco.mohfw.gov.in/sites/default/files/National_Data_Analysis_Plan_II_report.pdf
[15] Laga M. Effect of HIV prevention in key populations: evidence accumulates, time to implement. The Lancet Global Health, 2013; 1, e243-e24. [Online].; 2013 [cited 2025 Dec 31]. Accessed from: https://www.thelancet.com/journals/langlo/article/PIIS2214-109X(13)70096-2/fulltext
[16] Sathian B, Sreedharan J, Asim M, Menezes RG, van Teijlingen E, Unnikrishnan B. Estimation of the burden of people living with Human Immunodeficiency Virus/Acquired Immunodeficiency Syndrome (HIV/AIDS) in Kerala state, India. Nepal J Epidemiol. 2018 Sep 30;8(3):738-744. doi: 10.3126/nje. v8i3.23752. PMID: 31119040; PMCID: PMC6528742. [Online].; 2013 [cited 2025 Dec 31]. Accessed from: https://pmc.ncbi.nlm.nih.gov/articles/PMC6528742/
[17] Understanding measures of progress towards the 95–95–95 HIV testing, treatment, and viral suppression targets. The Joint United

Nations Programme on HIV/AIDS (UNAIDS) [Online].; 2023 [cited 2025 Dec 31]. Accessed from: https://www.unaids.org/sites/default/files/media_asset/progress-towards-95-95-95_en.pdf
[18] NACO. National Aids Control Organisation: HIV Facts & Figures. [Online].; 2024 [cited 2025 Dec 31]. Accessed from: http://naco.gov.in/hiv-facts-figures
[19] Tamil Nadu Aids Control Society. Celebrating 25 Years Journey in the Field of Prevention-to-Care Continuum of HIV / AIDS Services TNASC [Online]; 2022 [cited 2025 Dec 31]. Accessed from: https://tnsacs.in/cmsimage//IECMaterials/20012021050114-27.pdf.
[20] Solomon SS, Desai M, Srikrishnan AK, Thamburaj E, Vasudevan CK, Kumar MS, Solomon S, Celentano DD, Mehta SH. The profile of injection drug users in Chennai, India: identification of risk behaviors and implications for interventions. Subst Use Misuse. 2010; 45[3]:354-67. doi: 10.3109/10826080903452447. [Online]; 2010 [cited 2025 Dec 31]. Accessed from: https://pmc.ncbi.nlm.nih.gov/articles/PMC2924430/
[21] Isac S, Ramesh BM, Rajaram S, Washington R, Bradley JE, Reza-Paul S, Beattie TS, Alary M, Blanchard, JF, Moses S. Changes in HIV and syphilis prevalence among female sex workers from three serial cross-sectional surveys in Karnataka state, South India National Library of Medicine, BMJ Open. 2025; 5:3. [Online]; 2025 [cited 2025 Dec 31]. Accessed from: https://doi.org/10.1136/bmjopen-2014-007106

[22] Subramanian T, Ramakrishnan L, Aridoss S, Goswami P, Kanguswami B, Shajan M, Adhikary R, Purushothaman GK, Ramamoorthy SK, Chinnaswamy E, Veeramani IB, Paranjape RS. Increasing condom use and declining STI prevalence in high-risk MSM and TGs: evaluation of a large-scale prevention program in Tamil Nadu, India. BMC Public Health. 2013; 13:857. doi: 10.1186/1471-2458-13-857. [Online]; 2013 [cited 2025 Dec 31]. Accessed from: https://pubmed.ncbi.nlm.nih.gov/24044766/
[23] HIV/AIDS in India. [Online].; 2005 [cited 2025 Dec 31. Accessed from: https://www.aidsdatahub.org/sites/default/files/resource/hiv-aids-india.pdf
[24] TANSACS. Targeted Intervention. [Online]; 2010 [cited 2025 Dec 31]. Accessed from: https://tnsacs.in/page/109
[25] NACO. Assessment of Prevention of Parent to Child Transmission of HIV (PPTCT) Services Implementation in the states of Andhra Pradesh, Telengana, Karnataka, and Tamil Nadu, Assessment Report 2014. [Online].; 2014 [cited 2025 Dec 31]. Accessed from: https://naco.gov.in/sites/default/files/Final%20PPTCT%20assessment%20report%2020%2010%2014%20.pdf

[26] Santhakumar A, Nagaraj KJ, Malathi M, Ganesh B, Manikandan N, Padmapriya VM, Kanagasabai K, Elavarsu G, Ravi M, Nagaraju B, Prabakaran J, Kumar P, Rajan S, Elangovan A. Levels, and trend of HIV prevalence among pregnant women in Tamil Nadu: Analysis of data from HIV sentinel surveillance (2003–2019). Clin Epidemiol Glob Health. 2021; 9: 280-288. DOI: 10.1016/j.cegh.2020.09.011 [Online].; 2021 [cited 2025 Dec 31]. Accessed from: https://www.sciencedirect.com/science/article/pii/S2213398420302190

[27] NACO, PHFI: National AIDS Control Organization, Ministry of Health & Family Welfare, Govt. of India. Public Health Foundation of India (PHFI). Opioid Substitution Therapy Under National AIDS Control Programme: Clinical Practice Guidelines for Treatment with Buprenorphine. [Online].; 2021 [cited 2025 Dec 31]. Accessed from: https://www.lgbrimh.gov.in/resources/Addiction_Medicine/elibrary/opioid_substitution.pdf

[28] NACO. National AIDS Control Organization, Ministry of Health & Family Welfare, Govt. of India. Sankalak, Status of National AIDS Response, Second Edition 2020. [Online].; 2020 [cited 2025 Dec 31]. Accessed from: https://www.aidsdatahub.org/sites/default/files/resource/india-sankalak-report-second-edition.pdf

[29] NACO. National AIDS Control Organization, Ministry of Health & Family Welfare, Govt. of India: Injecting Drug Use: Strategic Report for NACP IV Planning. [Online].; 2015 [cited 2025 Dec 31]. Accessed from: https://www.naco.gov.in/sites/default/files/Strategy%20document%20Injecting%20Drug%20Use_final%20V2.pdf

[30] AIDSdataHub. AIDS Data hub for National AIDS Control Organization, Ministry of Health & Family Welfare, Govt. of India. INDIA HIV ESTIMATES 2021 Technical Brief, 2021. [Online].; 2021 [cited 2025 Dec 31]. Accessed from: https://www.aidsdatahub.org/sites/default/files/resource/india-hiv-estimates-2020-0.pdf

[31] NACO. National AIDS Control Organization, Ministry of Health & Family Welfare, Govt. of India: India HIV Estimates 2023 Technical Report. [Online].; 2023 [cited 2025 Dec 31]. Accessed from: https://naco.gov.in/sites/default/files/India%20HIV%20Estimates%202023_Technical%20Report_Final_17%20DEC%202024%20%281%29.pdf

[32] NEIGRIHMS. North Eastern Indira Gandhi Regional Institute of Health and Medical Sciences: HIV & AIDS POLICY -2023. [Online].; 2023 [cited 2025 Dec 31]. Accessed from: https://neigrihms.gov.in/Latest%20News/DDA/2023/Final%20HIV%20_%20AIDS%20Policy%202023.pdf

[33] Tamil Nadu's HIV Success: How 39-Year Fight Cut Infections by 85%? New Kerala.Com. [Online].; 2025. Accessed from: https://www.newkerala.com/news/o/cm-stalin-highlights-tns-remarkable-progress-reducing-hiv-prevalence-156

[34] NACO. National AIDS Control Organization, MOHFW, GOI: Strategy Document: National AIDS and STD Control Programme Phase-V (2021-2026) Anchoring the national response towards ending the AIDS epidemic. [Online].; 2023 [cited 2025 Dec 31]. Accessed from: https://naco.gov.in/sites/default/files/NACP_V_Strategy_Booklet.pdf
[35] NACO. National AIDS Control Organization, MOHFW, GOI: Assessment of ART Centers in India: A National Report; September 2015. [Online].; 2015 [cited 2025 Dec 31]. Accessed from: https://naco.gov.in/sites/default/files/ART%20Assessment%20National%20Report-Final%2028092015.pdf
[36] SIAI. Shankar IAS Academy Initiative: TNPSC Current Affairs: HIV/AIDS Prevalence - Tamil Nadu. [Online].; 2024 [cited 2025 Dec 31]. Accessed from: https://www.tnpscthervupettagam.com/currentaffairs-detail/hiv-aids-prevalence-%E2%80%93-tamilnadu?cat=tamilnadu-news
[37] World Bank, NACO: National AIDS Control Support Project, Independent Evaluation Group (IEG) Implementation Completion and Results Report (ICRR) Review. Ministry of Health and Family Welfare, Government of India; [Online].; 2021 [cited 2025 Dec 31]. Accessed from: https://naco.mohfw.gov.in/sites/default/files/IEG%20Implementation%20Completion%20and%20Result%20Report%20%20(1).pdf
[38] Hanenberg RS, Sokal DC, Rojanapithayakorn W, Kunasol P. Impact of Thailand's HIV-control programme as indicated by the decline of sexually transmitted diseases The Lancet 1994; 344, 243-245. DOI10.1016/s0140-6736(94)93004-x [Online].; 1994 [cited 2025 Dec 31]. Accessed from: https://pubmed.ncbi.nlm.nih.gov/7913163/
[39] Verma R. Scale-up and coverage of Avahan: a large-scale HIV-prevention programme among female sex workers and men who have sex with men in four Indian states. Sexually Transmitted Infections. 2010; doi:10.1136/STI.2009.039115 [Online].; 2010 [cited 2025 Dec 31]. Accessed from: https://sti.bmj.com/content/86/Suppl_1/i76
[40] UNAIDS. UNAIDS & AIDS Division, Ministry of Public Health, Thailand. Evaluation of the 100% Condom Programme in Thailand. A Case Study July 2000 [Online].; 2000 [cited 2025 Dec 31]. Accessed from: https://data.unaids.org/publications/irc-pub01/jc275-100pcondom_en.pdf
[41] Human Sciences Research Council (HSRC). The Sixth South African National HIV Prevalence, Incidence, and Behaviour Survey (SABSSM VI): Key findings for KwaZulu-Natal Province. Durban: HSRC; 2024. Available from: https://hsrc.ac.za/press-releases/phsb/kwazulu-natal-reports-second-highest-hiv-prevalence-rate-in-south-africa/
[42] Karim QA, Kharsany ABM, Leask K, Ntombela F, Humphries H, Frohlich JA, et al. Stabilizing HIV prevalence masks high HIV incidence rates amongst rural and urban women in KwaZulu-Natal, South Africa. Int J Epidemiol. 2011;40(4):922-30. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC3156366/

[43] Grulich AE, Guy R, Amin J, Jin F, Selvey C, Holden J, et al. Rapid reduction in HIV diagnoses after combined expansion of HIV testing and treatment and scaled-up antiretroviral pre-exposure prophylaxis in New South Wales, Australia: a prospective observational study. Lancet HIV. 2018;5(10): e552-61. Available from: https://doi.org/10.1016/S2352-3018(18)30215-7
[44] Schneider K, Gray RT, Wilson DP. A cost-effectiveness analysis of HIV pre-exposure prophylaxis for men who have sex with men in Australia. Med J Aust. 2014;201(11):653-6. Available from: https://pubmed.ncbi.nlm.nih.gov/24385445/
[45] Vassall A, Pickles M, Chandrashekar S et al. Cost-effectiveness of HIV prevention for high-risk groups at scale: an economic evaluation of the Avahan programme in south India. The Lancet Global Health, 2014; e531-e540. Available from: DOI: 10.1016/S2214-109X(14)70277-3.
[46] Bekker L, Alleyne G, Baral S et al. Advancing global health and strengthening the HIV response in the era of the Sustainable Development Goals: The International AIDS Society—Lancet Commission: The Lancet, 2018; 392(10144), 312-358 [Online].; 2024 [cited 2025 Dec 31]. Accessed from: https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(18)31070-5/fulltext
[47] Tran, N.T., Bennett, S.C., Bishnu, R. et al. Analyzing the sources and nature of influence: How the Avahan program used evidence to influence HIV/AIDS prevention policy in India. Implementation Sci, 2013; 8, 44. https://doi.org/10.1186/1748-5908-8-44
[48] Thilakavathi S, Boopathi K, Girish Kumar CP, Santhakumar A, Senthilkumar R, Eswaramurthy C, Ilaya Bharathy V, Ramakrishnan L, Thongamba G, Adhikary R, Paranjape R. Assessment of the scale, coverage and outcomes of the Avahan HIV prevention program for female sex workers in Tamil Nadu, India: is there evidence of an effect? BMC Public Health. 2011;11(Suppl 6): S3. doi: 10.1186/1471-2458-11-S6-S3. PMID: 22375609; PMCID: PMC3287556. [Online].; 2024 [cited 2025 Dec 31]. Accessed from: https://pubmed.ncbi.nlm.nih.gov/22375609/
[49] EKAM IAS Academy, AIDS And TB Fight — Tamil Nadu Shows the Way. Tamil Nadu's New TB Model [Online].; 2025 [cited 2025 Dec 31]. Accessed from: https://ekamiasacademy.in/aids-and-tb-fight-tamil-nadu-shows-the-way/
[50] Govt. of India (GOI}: National TB Elimination Program: India TB Report 2024. Central TB Division. Ministry of Health and Family Welfare. [Online].; 2024 [cited 2025 Dec 31]. Accessed from: https://tbcindia.nikshay.in/wp-content/uploads/2024/10/TB-Report_for-Web_08_10-2024-1.pdf
[51] Kumar P, Das C, Das U, Kumar A, Priyam N, Ranjan V, et al. (2023) Augmenting progress on the elimination of vertical transmissions of HIV in India: Insights from Spectrum-based HIV burden estimations. PLOS

Glob Public Health 3[8]: e0002270. https://doi.org/10.1371/journal.pgph.0002270
[52] National AIDS Control Organisation2023. Research Compendium Volume I from Research Studies undertaken by State AIDS Control Societies in FY 2021-22. New Delhi, NACO, Ministry of Health and Family Welfare, Government of India. Accessed from: https://naco.gov.in/sites/default/files/Final_Research%20Compendium%20Volume%201.pdf
[53] Grimes DA, Schulz KF. Descriptive studies: what they can and cannot do. Lancet. 2002;359(9301):145-9. Available from: https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(02)07376-9/fulltext
[54] Dahabreh IJ, Petito LC, Robertson SE, Hernán MA. Causally interpretable meta-analysis: Application in adolescent HIV prevention. Am J Epidemiol. 2020;189(11):1398-407. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC8742835/
[55] National AIDS Control Organisation (NACO). HIV Surveillance in India: Evolution and Changes. New Delhi: Ministry of Health and Family Welfare; 2010. Available from: https://www.aidsdatahub.org/resource/hiv-surveillance-india-evolution-and-changes-2010
[56] Green D, Tordoff DM, Kharono B, Akullian A, Bershteyn A, Morrison M, et al. Evidence of sociodemographic heterogeneity across the HIV treatment cascade and progress towards 90-90-90 in sub-Saharan Africa: a systematic review and meta-analysis. J Int AIDS Soc. 2020;23(3): e25470. Available from: https://onlinelibrary.wiley.com/doi/full/10.1002/jia2.25470
[57] Chimoyi L, Musekiwa A, Jaumdally S, et al. Challenges and barriers to HIV care engagement and care cascade: viewpoint. Front Public Health. 2023; 11:1159392. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC10398380/
[58] UNAIDS. Guidance for Triangulating Data and Estimates of HIV Prevalence Among Pregnant Women and Coverage of PMTCT Using the Spectrum AIDS Impact Module. Geneva: UNAIDS; 2023. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC11540279/
[59] National AIDS Control Organization. Ministry of Health and Family Welfare, Government of India. [Online].; 2017 [cited 2025 Dec 31. Accessed from: https://tnsacs.in/cmsimage//Guidelines/05012021080123.pdf
[60] Rai T, Lambert HS, Ward H. Complex routes into HIV care for migrant workers: a qualitative study from North India. AIDS Care. 2015;27(11):1418–1423. doi:10.1080/09540121.2015.1058898 [Online].; 2015 [cited 2025 Dec 31. Accessed from: https://pmc.ncbi.nlm.nih.gov/articles/PMC4706014/
[61] Subraminan Ma. Policy Note 2025-26. Demand No 19, by Minister for Health and Family Welfare, Health and Family Welfare Dept,

Government of Tamil Nadu. [Online].; 2025 [cited 2025 Dec 31. Accessed from: https://cms.tn.gov.in/cms_migrated/document/docfiles/hfw_e_pn_2025_26.pdf
[62] World Bank Group. HIV/AIDS in India. Feature Story July 10, 2012 HIV/AIDS in India [Online].; 2012. Accessed from: https://www.worldbank.org/en/news/feature/2012/07/10/hiv-aids-india

-o0o-